

# Notes on stability theory

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## Contents

|                                                                                                     |    |
|-----------------------------------------------------------------------------------------------------|----|
| Chapter 1. Stable formulas, both 2-valued and real-valued                                           | 5  |
| 1.1. The 2-valued case: $\varphi$ -types and type definability                                      | 5  |
| 1.2. The real-valued case                                                                           | 7  |
| 1.3. Historical remarks                                                                             | 13 |
| Chapter 2. Stability, trees, and definability of types                                              | 17 |
| 2.1. The equivalence proved in this chapter                                                         | 17 |
| 2.2. From the absence of the order property to type definability                                    | 18 |
| 2.3. From the order property to tall trees                                                          | 20 |
| 2.4. Tall trees produce non-definable types                                                         | 20 |
| 2.5. Type spaces over countable models: a perfect set dichotomy                                     | 22 |
| 2.6. Historical remarks                                                                             | 22 |
| Chapter 3. More stability, trees, and definability                                                  | 25 |
| 3.1. A quantitative combinatorial proof (using Ramsey's theorem) that trees give the order property | 25 |
| 3.2. Optional: The classical proof of definability, via ranks                                       | 27 |
| 3.3. Historical remarks                                                                             | 29 |
| Chapter 4. Stability and notions of independence                                                    | 31 |
| 4.1. Conventions                                                                                    | 31 |
| 4.2. Definitions and heirs                                                                          | 32 |
| 4.3. Independence relations                                                                         | 34 |
| 4.4. Remarks                                                                                        | 36 |
| 4.5. Historical remarks                                                                             | 36 |
| Chapter 5. From stability to NIP                                                                    | 39 |
| 5.1. The independence property                                                                      | 39 |
| 5.2. The real-valued case                                                                           | 40 |
| 5.3. Historical remarks                                                                             | 42 |
| Appendix A. Grothendieck's double limit criterion                                                   | 45 |
| A.1. The criterion                                                                                  | 45 |
| A.2. Historical remarks                                                                             | 47 |
| Bibliography                                                                                        | 49 |



## CHAPTER 1

### Stable formulas, both 2-valued and real-valued

Stability is a property of a single formula  $\varphi(x, y)$ , and this chapter looks at stability from two sides. On the logical side, a stable formula is one that cannot order arbitrarily long sequences of elements, and the central fact about stable formulas is that all their types are definable. In Section 1.1 we set up this side and state the definability theorem (see Theorem 1.1.6 below); the proof occupies Chapter 2. On the analytic side, a formula is simply a  $\{0, 1\}$ -valued function of two variables, and nothing prevents us from replacing such functions by bounded real-valued ones. In Section 1.2 we show that stability then becomes the statement that two iterated limits can be exchanged, a property closely tied to reflexive Banach spaces.

#### 1.1. The 2-valued case: $\varphi$ -types and type definability

Throughout this section and Chapters 2 and 3, we fix a first-order theory  $T$  in a language  $L$  and an  $L$ -formula  $\varphi(x, y)$  consistent with  $T$ , where  $x$  and  $y$  are finite tuples of variables. Tuples of elements are written  $a, b, \dots$  and are always of the appropriate length. We write  $\varphi^1 = \varphi$  and  $\varphi^0 = \neg\varphi$ .

**Definition 1.1.1.** Let  $M$  be a model of  $T$ .

- (a) A  $\varphi$ -formula over  $M$  is a formula  $\varphi(x, b)^u$  with  $b \in M$  and  $u \in \{0, 1\}$ .
- (b) A complete  $\varphi$ -type over  $M$  (in the variables  $x$ ) is a set  $p$  of  $\varphi$ -formulas over  $M$  such that
  - (i) the set  $p$  is consistent with the elementary diagram of  $M$ , and
  - (ii) for each  $b \in M$ , exactly one of  $\varphi(x, b)$  and  $\neg\varphi(x, b)$  belongs to  $p$ .

The set of complete  $\varphi$ -types over  $M$  will be denoted  $S_\varphi(M)$ .

If  $N \succeq M$  and  $c \in N$ , the set

$$\text{tp}_\varphi(c/M) = \{\varphi(x, b)^u : b \in M, u \in \{0, 1\}, N \models \varphi(c, b)^u\}$$

is a complete  $\varphi$ -type over  $M$ . Conversely, every complete  $\varphi$ -type  $p$  over  $M$  is consistent with the elementary diagram of  $M$ , so it is realized by some element  $c$  of some  $N \succeq M$ , and then  $p = \text{tp}_\varphi(c/M)$ .

For  $p \in S_\varphi(M)$  we write  $D_p = \{b \in M : \varphi(x, b) \in p\}$ . By condition (ii), we have  $p = \{\varphi(x, b) : b \in D_p\} \cup \{\neg\varphi(x, b) : b \in M \setminus D_p\}$ , so  $p$  is determined by  $D_p$ . The set  $D_p$  records, for each parameter  $b$ , whether  $\varphi(x, b)$  holds of the elements that realize  $p$ .

**Lemma 1.1.2 (finite satisfiability).** *Let  $p \in S_\varphi(M)$  be given. For every finite set  $\{\varphi(x, b_l)^{u_l} : l < k\} \subseteq p$  there is  $a \in M$  with  $M \models \bigwedge_{l < k} \varphi(a, b_l)^{u_l}$ .*

*Proof.* The  $L(M)$ -sentence  $\exists x \bigwedge_{l < k} \varphi(x, b_l)^{u_l}$  is consistent with the elementary diagram of  $M$ . That diagram is a complete  $L(M)$ -theory, so the sentence belongs to the diagram, that is, holds in  $M$ . ☺

**Definition 1.1.3.** Let  $p \in S_\varphi(M)$  be given.

- (a) The type  $p$  is *definable* if there is an  $L(M)$ -formula  $\psi(y)$ , a  $\varphi$ -*definition* of  $p$ , such that for all  $b \in M$ , we have

$$\varphi(x, b) \in p \iff M \models \psi(b).$$

Equivalently, the set  $D_p = \psi(M)$  is a definable subset of  $M$ .

- (b) A *positive Boolean combination of instances of  $\varphi$  over  $M$*  is a formula of the form

$$\bigvee_{i < n} \bigwedge_{a \in W_i} \varphi(a, y),$$

where  $n < \omega$  and  $W_0, \dots, W_{n-1}$  are finite subsets of  $M$ . By convention the empty disjunction is  $\perp$  and the empty conjunction is  $\top$ .

By distributivity, every formula built from formulas  $\varphi(a, y)$  ( $a \in M$ ) using only  $\wedge$  and  $\vee$  is equivalent to a positive Boolean combination of instances of  $\varphi$  over  $M$ . If  $\varphi$  is quantifier-free, so is every such combination.

In general  $D_p$  is just a subset of  $M$ , with no reason to be describable inside  $M$ . When  $M$  is infinite there are far more subsets of  $M$  than formulas over  $M$ , and a definable type is one whose set  $D_p$  is captured by a single formula. The motivating example is a realized type. If  $c \in M$ , then  $\text{tp}_\varphi(c/M)$  is defined by the formula  $\varphi(c, y)$ . So a definable type behaves like the type of an element of  $M$ , even when the type is realized only in an elementary extension.

The typical failure of definability comes from orders. In  $(\mathbb{Q}, <)$ , let  $\varphi(x, y)$  be  $x < y$  and let  $p$  be the type of  $\sqrt{2}$  over  $\mathbb{Q}$ . Then  $D_p = \{b \in \mathbb{Q} : b > \sqrt{2}\}$ , and  $p$  is not definable, since the subsets of  $\mathbb{Q}$  definable in  $(\mathbb{Q}, <)$  with parameters are finite unions of intervals with rational or infinite endpoints. By contrast, the type of an element just above a rational  $q$  is defined by  $y > q$ .

Definability also has a practical side. A formula  $\psi(y)$  makes sense in every elementary extension  $N \succeq M$ , so a definition of  $p$  tells us how to extend  $p$  to parameters in  $N$ : put  $\varphi(x, b)$  in the extension exactly when  $N \models \psi(b)$ . This is the source of the canonical extensions used to define independence in Chapter 4.

The example of  $(\mathbb{Q}, <)$  suggests that orders are the obstruction to definability. The next definition makes this idea precise.

**Definition 1.1.4.** Let  $M$  be a model of  $T$ , and let  $n \geq 1$  be given. An  *$n$ -ladder for  $\varphi$  in  $M$*  consists of tuples  $a_1, \dots, a_n$  and  $b_1, \dots, b_n$  of elements of  $M$  such that, for all  $1 \leq i, j \leq n$ , we have

$$M \models \varphi(a_i, b_j) \iff i \leq j.$$

We say that  $\varphi$  has the *order property* if for every  $n$  some model of  $T$  contains an  $n$ -ladder for  $\varphi$ , and that  $\varphi$  is *stable* otherwise.

Figure 1.1 explains the name. Put the  $a_i$  on one rail and the  $b_j$  on the other, both in increasing order, and join  $a_i$  to  $b_j$  whenever  $\varphi(a_i, b_j)$  holds. The “horizontal” pairs  $(a_i, b_i)$  form the rungs of a ladder. The defining feature is the asymmetry between the two directions: each  $a_i$  is related to every  $b_j$  with  $j \geq i$  and to none with  $j < i$ . So  $\varphi$  linearly orders the rungs, in the sense that  $i \leq j$  if and only if  $\varphi(a_i, b_j)$ .



**Remark 1.1.5.** An  $(n+1)$ -ladder contains an  $n$ -ladder. Hence, by compactness, the formula  $\varphi$  has the order property if and only if some model of  $T$  contains an infinite sequence  $(a_i, b_i)_{i < \omega}$  with  $\varphi(a_i, b_j) \iff i \leq j$ . The choice of “ $\leq$ ” is immaterial. Suppose that  $a_1, \dots, a_n$  and  $b_1, \dots, b_n$  are tuples in a model of  $T$  such that  $\varphi(a_i, b_j) \iff i > j$  for  $1 \leq i, j \leq n$ . Reversing the indices by putting  $a'_i = a_{n+1-i}$  and  $b'_j = b_{n+1-j}$  gives  $\varphi(a'_i, b'_j) \iff i < j$ . Shifting by putting  $b''_j = b'_{j+1}$  then gives  $\varphi(a'_i, b''_j) \iff i \leq j$  for  $1 \leq i, j \leq n-1$ . So a “ $>$ ”-pattern of length  $n$  yields an  $(n-1)$ -ladder.

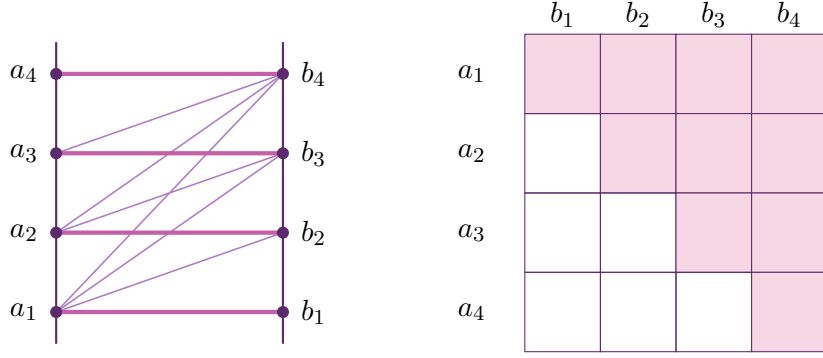


FIGURE 1.1. A 4-ladder. Left: the elements  $a_i$  and  $b_j$  are joined when  $\varphi(a_i, b_j)$  holds, i.e. when  $i \leq j$ . The pairs  $(a_i, b_i)$  are the rungs (thick, pink), and each  $a_i$  also reaches every  $b_j$  with  $j > i$  (purple). Right: the truth table of  $\varphi(a_i, b_j)$ , with pink cells where  $\varphi$  holds. The table is a staircase, which is why an  $n$ -ladder is also called a *half-graph*.

The main theorem of this chapter confirms that the order property is the only obstruction to definability. Moreover, when there is no obstruction, the definitions can be chosen of the simplest possible kind, namely positive Boolean combinations of instances of  $\varphi$  (see Definition 1.1.3(b)).

**Theorem 1.1.6 (Definability of types).** *The following are equivalent.*

- (1) *The formula  $\varphi$  does not have the order property.*
- (2) *For every model  $M$  of  $T$ , every  $p \in S_\varphi(M)$  has a  $\varphi$ -definition that is a positive Boolean combination of instances  $\varphi(a, y)$  with  $a \in M$ .*
- (3) *For every model  $M$  of  $T$ , every  $p \in S_\varphi(M)$  is definable.*

In short: *all  $\varphi$ -types are definable if and only if  $\varphi$  does not have the order property.* The implication  $(1) \Rightarrow (2)$  is proved in Section 2.2, and  $(2) \Rightarrow (3)$  is trivial. For  $(3) \Rightarrow (1)$  we show in Sections 2.3–2.4 that if  $\varphi$  has the order property, then some  $\varphi$ -type is not definable. The route passes through binary trees.

## 1.2. The real-valued case

In this section we follow Iovino [49]. A formula is identified with a  $\{0, 1\}$ -valued function, and stability becomes a property of bounded real-valued functions: the exchangeability of iterated limits. This point of view connects stability with reflexivity of Banach spaces, via a characterization of reflexivity due to James [50] that closely resembles the order property.

**1.2.1. Stability and iterated limit exchange.** Let  $M$  be an  $L$ -structure and  $\varphi(x, y)$  an  $L$ -formula. We identify  $\varphi$  with the function  $\varphi: M^{|x|} \times M^{|y|} \rightarrow \{0, 1\}$  that takes the value 1 at  $(a, b)$  if and only if  $M \models \varphi(a, b)$ . More generally, a function  $\varphi: A \times B \rightarrow \{0, 1\}$  on arbitrary sets  $A, B$  has the *order property* if there are sequences  $(a_m)_{m < \omega}$  in  $A$  and  $(b_n)_{n < \omega}$  in  $B$  such that

$$\varphi(a_m, b_n) = 1 \iff m \leq n.$$

**Definition 1.2.1.** A function  $\varphi: A \times B \rightarrow \{0, 1\}$  is *unstable* if either  $\varphi$  or  $\neg\varphi = 1 - \varphi$  has the order property, and *stable* otherwise. A formula  $\varphi(x, y)$  is *stable on  $M$*  if the corresponding function on  $M^{|x|} \times M^{|y|}$  is stable.



**Remark 1.2.2.** A formula  $\varphi(x, y)$  is stable in the sense of Definition 1.1.4 if and only if  $\varphi$  is stable on every model of  $T$ . Indeed, if  $\varphi$  has the order property, then by compactness (see Remark 1.1.5) some model contains an infinite ladder, so  $\varphi$  is unstable on that model. Conversely, if  $\varphi$  or  $\neg\varphi$  has the order property on some model  $M$  of  $T$ , then  $M$  contains  $n$ -ladders for  $\varphi$  for every  $n$ : directly in the first case, and by Remark 1.1.5 in the second, since  $\neg\varphi(a_m, b_n) \iff m \leq n$  means  $\varphi(a_m, b_n) \iff m > n$ .

To state the analytic form of stability, we use limits along ultrafilters. For a family  $(a_i)_{i \in I}$  in a topological space  $X$ , a point  $a \in X$ , and an ultrafilter  $\mathcal{U}$  on  $I$ , we write

$$(1.1) \quad \lim_{i, \mathcal{U}} a_i = a$$

if  $a$  is a limit of  $(a_i)$  along  $\mathcal{U}$ . We use freely the standard facts about such limits: limits along ultrafilters are unique in Hausdorff spaces, always exist in compact spaces, describe closures, and are preserved by continuous maps.

The next proposition, which we use several times, extracts sequences from iterated limits taken along arbitrary families.

**Proposition 1.2.3.** Let  $\varphi: A \times B \rightarrow [0, 1]$  be a function, let  $(a_i)_{i \in I}$  be a family in  $A$  and  $(b_j)_{j \in J}$  a family in  $B$ , and let  $\mathcal{U}, \mathcal{V}$  be ultrafilters on  $I, J$  with

$$\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \alpha, \quad \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j) = \beta.$$

Then there are  $i_0, i_1, \dots \in I$  and  $j_0, j_1, \dots \in J$  such that

$$|\varphi(a_{i_m}, b_{j_n}) - \alpha| \leq 2^{1-m} \text{ if } m < n, \quad |\varphi(a_{i_m}, b_{j_n}) - \beta| \leq 2^{1-n} \text{ if } n < m.$$

*Proof.* By definition of the limits, for every  $\varepsilon > 0$  there are  $U_\varepsilon \in \mathcal{U}$  and  $V_\varepsilon \in \mathcal{V}$  such that

$$\begin{aligned} i \in U_\varepsilon &\implies \left| \lim_{j, \mathcal{V}} \varphi(a_i, b_j) - \alpha \right| \leq \varepsilon, \\ j \in V_\varepsilon &\implies \left| \lim_{i, \mathcal{U}} \varphi(a_i, b_j) - \beta \right| \leq \varepsilon. \end{aligned}$$

Also, for all  $\hat{i} \in I, \hat{j} \in J$  and  $\varepsilon > 0$  there are  $V_\varepsilon^{\hat{i}} \in \mathcal{V}$  and  $U_\varepsilon^{\hat{j}} \in \mathcal{U}$  such that

$$\begin{aligned} j \in V_\varepsilon^{\hat{i}} &\implies \left| \varphi(a_{\hat{i}}, b_j) - \lim_{j, \mathcal{V}} \varphi(a_{\hat{i}}, b_j) \right| \leq \varepsilon, \\ i \in U_\varepsilon^{\hat{j}} &\implies \left| \varphi(a_i, b_{\hat{j}}) - \lim_{i, \mathcal{U}} \varphi(a_i, b_{\hat{j}}) \right| \leq \varepsilon. \end{aligned}$$

Choose recursively

$$i_N \in U_{2^{-N}} \cap \bigcap_{k < N} U_{2^{-k}}^{j_k}, \quad j_N \in V_{2^{-N}} \cap \bigcap_{k \leq N} V_{2^{-k}}^{i_k}.$$

These sets are finite intersections of members of an ultrafilter, hence nonempty. If  $m < n$ , then  $j_n \in V_{2^{-n}}^{i_m}$  and  $i_m \in U_{2^{-m}}$ , so  $|\varphi(a_{i_m}, b_{j_n}) - \alpha| \leq 2^{-n} + 2^{-m} \leq 2^{1-m}$ . If  $n < m$ , then  $i_m \in U_{2^{-m}}^{j_n}$  and  $j_n \in V_{2^{-n}}$ , so  $|\varphi(a_{i_m}, b_{j_n}) - \beta| \leq 2^{-m} + 2^{-n} \leq 2^{1-n}$ . ☺

**Proposition 1.2.4.** Let  $\varphi: A \times B \rightarrow \{0, 1\}$  be a function. The following are equivalent.

- (1) The function  $\varphi$  is stable.
- (2) For all sequences  $(a_m)$  in  $A$  and  $(b_n)$  in  $B$  and all ultrafilters  $\mathcal{U}, \mathcal{V}$  on  $\mathbb{N}$ , we have

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n).$$



*Proof.* (2)  $\Rightarrow$  (1). Suppose that  $\varphi$  is unstable. If  $\varphi$  has the order property, there are sequences  $(a_m)$  in  $A$  and  $(b_n)$  in  $B$  with  $\varphi(a_m, b_n) = 1 \iff m \leq n$ . Let  $\mathcal{U} = \mathcal{V}$  be a nonprincipal ultrafilter on  $\mathbb{N}$ . For fixed  $m$  we have  $\varphi(a_m, b_n) = 1$  for all  $n \geq m$ , so the left-hand side of (2) is 1. For fixed  $n$  we have  $\varphi(a_m, b_n) = 0$  for all  $m > n$ , so the right-hand side is 0. If instead  $\neg\varphi$  has the order property, the two sides are 0 and 1.

(1)  $\Rightarrow$  (2). Suppose that (2) fails: for some sequences  $(a_m)$ ,  $(b_n)$  and ultrafilters  $\mathcal{U}$ ,  $\mathcal{V}$ , the two iterated limits are  $\alpha \neq \beta$ . Since  $\varphi$  is  $\{0, 1\}$ -valued, both limits lie in  $\{0, 1\}$ . By Proposition 1.2.3 there are sequences  $(a'_m)$  in  $A$  and  $(b'_n)$  in  $B$  with  $|\varphi(a'_m, b'_n) - \alpha| \leq 2^{1-m}$  for  $m < n$  and  $|\varphi(a'_m, b'_n) - \beta| \leq 2^{1-n}$  for  $n < m$ . Since  $\varphi$  is  $\{0, 1\}$ -valued, we have  $\varphi(a'_m, b'_n) = \alpha$  for  $2 \leq m < n$  and  $\varphi(a'_m, b'_n) = \beta$  for  $2 \leq n < m$ . Put  $c_m = a'_{2m+2}$  and  $d_n = b'_{2n+3}$ . Since  $2m+2 < 2n+3 \iff m \leq n$ , we get  $\varphi(c_m, d_n) = \alpha$  if  $m \leq n$  and  $\varphi(c_m, d_n) = \beta$  if  $m > n$ . So  $\varphi$  has the order property if  $\alpha = 1$ , and  $\neg\varphi$  does if  $\alpha = 0$ .  $\odot$

Condition (2) of Proposition 1.2.4 makes sense for every bounded real-valued function, and we take this condition as the definition of stability in the real-valued setting.

**Definition 1.2.5.** A bounded function  $\varphi: A \times B \rightarrow \mathbb{R}$  is *stable* if, for all sequences  $(a_m)$  in  $A$  and  $(b_n)$  in  $B$  and all ultrafilters  $\mathcal{U}, \mathcal{V}$  on  $\mathbb{N}$ , we have

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n).$$

By Proposition 1.2.4, for  $\{0, 1\}$ -valued functions this agrees with Definition 1.2.1. By Proposition 1.2.3, the sequences in Definition 1.2.5 can be replaced by arbitrary families. For a bounded set of continuous functions on a compact space, the same condition is the hypothesis of Grothendieck's double limit criterion (see Appendix A below).

**Corollary 1.2.6.** If  $\varphi: A \times B \rightarrow [0, 1]$  is stable, then  $\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j)$  for all families  $(a_i)_{i \in I}$ ,  $(b_j)_{j \in J}$  and all ultrafilters  $\mathcal{U}$  on  $I$ ,  $\mathcal{V}$  on  $J$ .

*Proof.* If the two limits are  $\alpha \neq \beta$ , take  $(i_m)$ ,  $(j_n)$  as in Proposition 1.2.3 and a nonprincipal ultrafilter  $\mathcal{W}$  on  $\mathbb{N}$ . For fixed  $m$ , the limit  $\lim_{n, \mathcal{W}} \varphi(a_{i_m}, b_{j_n})$  is within  $2^{1-m}$  of  $\alpha$ , so  $\lim_{m, \mathcal{W}} \lim_{n, \mathcal{W}} \varphi(a_{i_m}, b_{j_n}) = \alpha$ . Symmetrically, the other iterated limit is  $\beta$ , which contradicts the stability of  $\varphi$ .  $\odot$

The proof of Corollary 1.2.6 used a single nonprincipal ultrafilter  $\mathcal{W}$  on  $\mathbb{N}$ , in both coordinates, and this is no accident. For ultrafilters  $\mathcal{U}$  on  $I$  and  $\mathcal{V}$  on  $J$ , let  $\mathcal{U} \otimes \mathcal{V}$  be the ultrafilter on  $I \times J$  consisting of the sets  $X$  with  $\{i : \{j : (i, j) \in X\} \in \mathcal{V}\} \in \mathcal{U}$ . For every bounded  $f: I \times J \rightarrow \mathbb{R}$  we have  $\lim_{(i, j), \mathcal{U} \otimes \mathcal{V}} f(i, j) = \lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} f(i, j)$ , so the two iterated limits of Definition 1.2.5 are the limits of  $\varphi(a_m, b_n)$  along  $\mathcal{U} \otimes \mathcal{V}$  and along the transpose of  $\mathcal{V} \otimes \mathcal{U}$ . The tensor product of ultrafilters is not commutative. If  $\mathcal{W}$  is a nonprincipal ultrafilter on  $\mathbb{N}$ , then the upper triangle  $\{(m, n) : m < n\}$  belongs to  $\mathcal{W} \otimes \mathcal{W}$ , because each of its rows is cofinite, while the lower triangle  $\{(m, n) : m > n\}$  does not, because each of its rows is finite. A ladder is exactly a pair of sequences along which  $\varphi$  is the indicator function of the closed upper triangle  $\{(m, n) : m \leq n\}$ . So the order property says that  $\varphi$  can tell  $\mathcal{W} \otimes \mathcal{W}$  from its own transpose, and the proofs of Proposition 1.2.4 and Corollary 1.2.6 show the following: a bounded function  $\varphi$  is stable if and only if, for one (equivalently, for every) nonprincipal ultrafilter  $\mathcal{W}$  on  $\mathbb{N}$  and all sequences  $(a_m)$  and  $(b_n)$ , we have

$$\lim_{m, \mathcal{W}} \lim_{n, \mathcal{W}} \varphi(a_m, b_n) = \lim_{n, \mathcal{W}} \lim_{m, \mathcal{W}} \varphi(a_m, b_n).$$

Stable functions are blind to the noncommutativity of the tensor square of a single ultrafilter.

**1.2.2. Stability and continuous extendability.** In this subsection we show that a function  $\varphi: A \times B \rightarrow [0, 1]$  is stable exactly when  $\varphi$  extends to a separately continuous function on pairs of types (see Proposition 1.2.8 below). We first define the spaces of types.

**Definition 1.2.7.** Let  $\varphi: A \times B \rightarrow [0, 1]$  be a function. For  $a \in A$ , the *left  $\varphi$ -type* of  $a$  is the function  $\text{lt}_\varphi(a): y \mapsto \varphi(a, y)$  on  $B$ . For  $b \in B$ , the *right  $\varphi$ -type* of  $b$  is the function  $\text{rt}_\varphi(b): x \mapsto \varphi(x, b)$  on  $A$ . The *space of left  $\varphi$ -types*  $\text{LS}(\varphi)$  is the closure of  $\{\text{lt}_\varphi(a) : a \in A\}$  in  $[0, 1]^B$  with the product topology, and the *space of right  $\varphi$ -types*  $\text{RS}(\varphi)$  is the closure of  $\{\text{rt}_\varphi(b) : b \in B\}$  in  $[0, 1]^A$ .

By Tychonoff's theorem, the spaces  $[0, 1]^A$  and  $[0, 1]^B$  are compact, and so are their closed subspaces  $\text{RS}(\varphi)$  and  $\text{LS}(\varphi)$ . Both spaces are quotients of Stone-Ćech compactifications. The map  $a \mapsto \text{lt}_\varphi(a)$  from the discrete space  $A$  into the compact space  $[0, 1]^B$  extends to a continuous map  $\beta A \rightarrow [0, 1]^B$ , namely  $\mathcal{U} \mapsto \lim_{a, \mathcal{U}} \text{lt}_\varphi(a)$ , whose image is compact and contains the realized types, and is therefore  $\text{LS}(\varphi)$ . So every left  $\varphi$ -type is the limit of the realized types along an ultrafilter on  $A$ , the principal ultrafilters give the realized types, and  $\text{LS}(\varphi)$  is a continuous image of  $\beta A$ ; likewise  $\text{RS}(\varphi)$  is a continuous image of  $\beta B$ .

**Proposition 1.2.8.** For  $\varphi: A \times B \rightarrow [0, 1]$  the following are equivalent.

- (1) The function  $\varphi$  is stable.
- (2) There is a separately continuous function  $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$  such that

$$\hat{\varphi}(\text{lt}_\varphi(a), \text{rt}_\varphi(b)) = \varphi(a, b) \quad \text{for all } (a, b) \in A \times B.$$

*Proof.* (1)  $\Rightarrow$  (2). Let  $p \in \text{LS}(\varphi)$  and  $q \in \text{RS}(\varphi)$  be given. There are families  $(a_i)_{i \in I}$  in  $A$ ,  $(b_j)_{j \in J}$  in  $B$  and ultrafilters  $\mathcal{U}$  on  $I$ ,  $\mathcal{V}$  on  $J$  with

$$p = \lim_{i, \mathcal{U}} \text{lt}_\varphi(a_i), \quad q = \lim_{j, \mathcal{V}} \text{rt}_\varphi(b_j).$$

By Corollary 1.2.6, we have  $\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j)$ . Define  $\hat{\varphi}(p, q)$  to be this common value. Since limits in the product topology are computed coordinatewise, this common value satisfies

$$(1.2) \quad \hat{\varphi}(p, q) = \lim_{i, \mathcal{U}} q(a_i) = \lim_{j, \mathcal{V}} p(b_j).$$

$\hat{\varphi}$  is well defined. Suppose also  $p = \lim_{i, \mathcal{U}'} \text{lt}_\varphi(a'_i)$  and  $q = \lim_{j, \mathcal{V}'} \text{rt}_\varphi(b'_j)$ . Then  $\lim_{j, \mathcal{V}} \varphi(x, b_j) = q(x) = \lim_{j, \mathcal{V}'} \varphi(x, b'_j)$  for every  $x \in A$ , and  $\lim_{i, \mathcal{U}} \varphi(a_i, y) = p(y) = \lim_{i, \mathcal{U}'} \varphi(a'_i, y)$  for every  $y \in B$ . Hence, using Corollary 1.2.6 for the middle step, we have

$$\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}'} \varphi(a_i, b'_j) = \lim_{j, \mathcal{V}'} \lim_{i, \mathcal{U}} \varphi(a_i, b'_j) = \lim_{j, \mathcal{V}'} \lim_{i, \mathcal{U}'} \varphi(a'_i, b'_j).$$

$\hat{\varphi}$  extends  $\varphi$ . Take constant families.

$\hat{\varphi}$  is separately continuous. Fix  $p$ , and let  $q = \lim_{k, \mathcal{W}} q_k$  in  $\text{RS}(\varphi)$ , where  $\mathcal{W}$  is an ultrafilter on a set  $K$ . Write  $q_k = \lim_{j, \mathcal{V}_k} \text{rt}_\varphi(b_{k,j})$  with  $j$  ranging over a set  $J_k$ . On  $J' = \{(k, j) : k \in K, j \in J_k\}$  let  $\mathcal{V}'$  be the ultrafilter of all  $X \subseteq J'$  with  $\{k : \{j : (k, j) \in X\} \in \mathcal{V}_k\} \in \mathcal{W}$ . For every family  $(z_{k,j})$  in a compact Hausdorff space we have  $\lim_{(k,j), \mathcal{V}'} z_{k,j} = \lim_{k, \mathcal{W}} \lim_{j, \mathcal{V}_k} z_{k,j}$ . Hence  $q = \lim_{(k,j), \mathcal{V}'} \text{rt}_\varphi(b_{k,j})$ , and by (1.2), we have

$$\hat{\varphi}(p, q) = \lim_{(k,j), \mathcal{V}'} p(b_{k,j}) = \lim_{k, \mathcal{W}} \lim_{j, \mathcal{V}_k} p(b_{k,j}) = \lim_{k, \mathcal{W}} \hat{\varphi}(p, q_k).$$

So  $\hat{\varphi}(p, \cdot)$  is continuous. Continuity of  $\hat{\varphi}(\cdot, q)$  is proved symmetrically.

(2)  $\Rightarrow$  (1). Let  $(a_m)$ ,  $(b_n)$  be sequences in  $A$ ,  $B$ , let  $\mathcal{U}, \mathcal{V}$  be ultrafilters on  $\mathbb{N}$ , and put  $p = \lim_{m, \mathcal{U}} \text{lt}_\varphi(a_m)$  and  $q = \lim_{n, \mathcal{V}} \text{rt}_\varphi(b_n)$ . Then  $(p, q) \in \text{LS}(\varphi) \times \text{RS}(\varphi)$ , and by separate

continuity

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{m, \mathcal{U}} \hat{\varphi}(\text{ltp}_\varphi(a_m), q) = \hat{\varphi}(p, q) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n). \quad \text{☺}$$

**1.2.3. Stability and reflexivity.** In this subsection we state Iovino's characterization of the stable functions as restrictions of the pairing between the unit balls of a reflexive Banach space  $E$  and of  $E^*$  (see Theorem 1.2.9 below).

For a Banach space  $E$  with dual  $E^*$ , we write  $B_E$  for the closed unit ball of  $E$  and  $\langle x, y \rangle$  for  $y(x)$ , where  $x \in E$  and  $y \in E^*$ . Recall that  $E$  is *reflexive* if the canonical embedding of  $E$  into  $E^{**}$  is surjective.

**Theorem 1.2.9 (Iovino).** *For  $\varphi: A \times B \rightarrow [0, 1]$  the following are equivalent.*

- (1) *The function  $\varphi$  is stable.*
- (2) *There are a reflexive Banach space  $E$  and maps  $u: A \rightarrow B_E$  and  $v: B \rightarrow B_{E^*}$  such that*

$$\varphi(x, y) = \langle u(x), v(y) \rangle \quad \text{for all } (x, y) \in A \times B.$$

*Proof.* See [49, Theorem 3.3]. For (1)  $\Rightarrow$  (2), stability is used to show that the operator  $T: \ell^\infty(A)^* \rightarrow \ell^\infty(B)$  given by  $T\mu(y) = \int \varphi(x, y) d\mu(x)$  is weakly compact; this step goes back to Krivine and Maurey [51] and Raynaud [52]. The theorem of Davis, Figiel, Johnson and Pełczyński [46] then factors  $T$  through a reflexive space. The implication (2)  $\Rightarrow$  (1) follows from the weak compactness of  $B_E$  and the weak\* compactness of  $B_{E^*}$ . ☺

By Remark 1.2.2 and Proposition 1.2.4, a formula  $\varphi(x, y)$  is stable in the sense of Definition 1.1.4 if and only if, in every model  $M$  of  $T$ , the function  $\varphi$  on  $M$  is the restriction of the pairing map  $B_E \times B_{E^*} \rightarrow \mathbb{R}$  for some reflexive Banach space  $E$  (depending on  $M$ ), composed with maps  $M^{|x|} \rightarrow B_E$  and  $M^{|y|} \rightarrow B_{E^*}$ .

**1.2.4. Stability and polynomial approximation.** By Theorem 1.1.6, if  $\varphi$  does not have the order property, every  $\varphi$ -type over a model of  $T$  is defined by a Boolean combination of instances  $\varphi(a, y)$ . On  $\{0, 1\}$  the Boolean connectives are polynomials, namely  $\neg t = 1 - t$  and  $t \wedge s = ts$ . So the real-valued analogue of a definition is a *uniform limit of polynomials in instances of  $\varphi$* . In this subsection we use the Stone–Weierstrass theorem to show that stability is exactly what is needed for such approximations to exist (see Theorem 1.2.11 below). Continuous logic treats definability in the same way, and Ben Yaacov used polynomial approximation to derive definability of types from Grothendieck's double limit criterion (see the historical remarks at the end of this chapter, and Appendix A). We use the Stone–Weierstrass theorem in the following form.

**Fact 1.2.10 (Stone–Weierstrass).** *Let  $K$  be a compact Hausdorff space and let  $\mathcal{F} \subseteq C(K)$  be a family of continuous real-valued functions that separates the points of  $K$ . Then every  $f \in C(K)$  is a uniform limit on  $K$  of functions of the form  $P(g_1, \dots, g_k)$ , where  $g_1, \dots, g_k \in \mathcal{F}$  and  $P$  is a real polynomial in  $k$  variables.*

Fix  $\varphi: A \times B \rightarrow [0, 1]$ . For  $a \in A$ , let  $e_a: \text{RS}(\varphi) \rightarrow [0, 1]$  be the evaluation map  $e_a(q) = q(a)$ , and for  $b \in B$ , let  $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$  be  $e_b(p) = p(b)$ . These maps are continuous, because  $\text{RS}(\varphi)$  and  $\text{LS}(\varphi)$  carry the product topology. On realized types, these maps are instances of  $\varphi$ , since  $e_a(\text{rtp}_\varphi(b)) = \varphi(a, b) = e_b(\text{ltp}_\varphi(a))$ . The family  $\{e_a : a \in A\}$  separates the points of  $\text{RS}(\varphi)$ , since two distinct functions  $q \neq q'$  in  $[0, 1]^A$  differ at some  $a \in A$ . Likewise  $\{e_b : b \in B\}$  separates the points of  $\text{LS}(\varphi)$ .

**Theorem 1.2.11.** *For  $\varphi: A \times B \rightarrow [0, 1]$  the following are equivalent.*

- (1) *The function  $\varphi$  is stable.*

(2) *There is a function  $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$  with  $\hat{\varphi}(\text{ltp}_\varphi(a), \text{rtp}_\varphi(b)) = \varphi(a, b)$  for all  $(a, b) \in A \times B$ , such that for every  $\varepsilon > 0$ :*

(a) *for every  $p \in \text{LS}(\varphi)$  there are  $a_1, \dots, a_k \in A$  and a real polynomial  $P$  with*

$$|\hat{\varphi}(p, q) - P(q(a_1), \dots, q(a_k))| \leq \varepsilon \quad \text{for all } q \in \text{RS}(\varphi);$$

(b) *for every  $q \in \text{RS}(\varphi)$  there are  $b_1, \dots, b_l \in B$  and a real polynomial  $Q$  with*

$$|\hat{\varphi}(p, q) - Q(p(b_1), \dots, p(b_l))| \leq \varepsilon \quad \text{for all } p \in \text{LS}(\varphi).$$

*Proof.* (1)  $\Rightarrow$  (2). Let  $\hat{\varphi}$  be the separately continuous function given by Proposition 1.2.8. For fixed  $p$ , the function  $\hat{\varphi}(p, \cdot)$  is continuous on the compact Hausdorff space  $\text{RS}(\varphi)$ . Applying Fact 1.2.10 with  $\mathcal{F} = \{e_a : a \in A\}$  gives (a). Clause (b) follows in the same way, using the continuity of  $\hat{\varphi}(\cdot, q)$  on  $\text{LS}(\varphi)$  and  $\mathcal{F} = \{e_b : b \in B\}$ .

(2)  $\Rightarrow$  (1). Each function  $q \mapsto P(q(a_1), \dots, q(a_k))$  is continuous on  $\text{RS}(\varphi)$ . By (a), the function  $\hat{\varphi}(p, \cdot)$  is a uniform limit of such functions, hence continuous. By (b), the function  $\hat{\varphi}(\cdot, q)$  is continuous for the same reason. Hence  $\hat{\varphi}$  is separately continuous, and  $\varphi$  is stable by Proposition 1.2.8.  $\text{☺}$

Suppose that  $\varphi$  is stable, and let  $\hat{\varphi}$  be the separately continuous extension given by Proposition 1.2.8, as in the proof of (1)  $\Rightarrow$  (2). Then  $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = p(b)$  for  $p \in \text{LS}(\varphi)$  and  $b \in B$ : if  $p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i)$ , continuity in the first variable gives  $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = \lim_{i, \mathcal{U}} \varphi(a_i, b) = p(b)$ . Taking  $q = \text{rtp}_\varphi(b)$  in (a) therefore gives

$$|p(b) - P(\varphi(a_1, b), \dots, \varphi(a_k, b))| \leq \varepsilon \quad \text{for all } b \in B.$$

So every left  $\varphi$ -type is a uniform limit on  $B$  of polynomials in instances  $\varphi(a, y)$  with  $a \in A$ , and symmetrically for right types.

The approximation in Theorem 1.2.11 is required on all of  $\text{RS}(\varphi)$  (respectively  $\text{LS}(\varphi)$ ), not only at realized types. The next example shows that approximation on  $B$  alone does not imply stability.

**Example 1.2.12.** Let  $A = B = \mathbb{N}$  and  $\varphi(m, n) = 1$  if  $m \leq n$ ,  $\varphi(m, n) = 0$  otherwise. Then  $\varphi$  has the order property and is therefore unstable. Here  $\text{ltp}_\varphi(m)$  is the indicator function of  $[m, \infty)$ , and  $\text{RS}(\varphi)$  consists of the indicator functions of the intervals  $[0, n]$  together with the constant function  $\mathbf{1}$ . Similarly, the space  $\text{LS}(\varphi)$  consists of the functions  $\text{ltp}_\varphi(m)$  together with the constant function  $\mathbf{0}$ . So every  $p \in \text{LS}(\varphi)$ , viewed as a function on  $B$ , is either an instance  $\varphi(m, \cdot)$  or the constant 0, which is trivially a polynomial in instances. Symmetrically, every  $q \in \text{RS}(\varphi)$  is an instance or the constant 1. The failure appears at the unrealized pair  $(\mathbf{0}, \mathbf{1})$ . If  $\hat{\varphi}$  were separately continuous and extended  $\varphi$  as in Theorem 1.2.11, then  $\hat{\varphi}(\mathbf{0}, \text{rtp}_\varphi(n)) = \mathbf{0}(n) = 0$  for all  $n$ , and  $\text{rtp}_\varphi(n) \rightarrow \mathbf{1}$ , so  $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 0$ . But also  $\hat{\varphi}(\text{ltp}_\varphi(m), \mathbf{1}) = \mathbf{1}(m) = 1$  for all  $m$ , and  $\text{ltp}_\varphi(m) \rightarrow \mathbf{0}$ , so  $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 1$ .

**Corollary 1.2.13.** *Let  $M$  be an  $L$ -structure and let  $\varphi(x, y)$  be a formula that is stable on  $M$ . Then every  $p \in \text{LS}(\varphi)$  is defined by a Boolean combination of instances: there are  $a_1, \dots, a_k \in M^{|x|}$  and a Boolean combination  $\psi(y)$  of  $\varphi(a_1, y), \dots, \varphi(a_k, y)$  such that, for all  $b \in M^{|y|}$ , we have*

$$p(b) = 1 \iff M \models \psi(b).$$

*Proof.* Apply Theorem 1.2.11 with  $\varepsilon = \frac{1}{3}$  to get  $a_1, \dots, a_k$  and  $P$ . Both  $p(b)$  and  $\varphi(a_i, b)$  lie in  $\{0, 1\}$ , so  $p(b) = 1$  if and only if  $t = (\varphi(a_1, b), \dots, \varphi(a_k, b))$  belongs to  $S = \{t \in \{0, 1\}^k : P(t) > \frac{1}{2}\}$ . Take  $\psi(y) = \bigvee_{t \in S} \bigwedge_{i \leq k} \varphi(a_i, y)^{t_i}$  (recall that  $\varphi^1 = \varphi$  and  $\varphi^0 = \neg\varphi$ ).  $\text{☺}$

Now let  $M$  be a model of  $T$ , and identify each  $p \in S_\varphi(M)$  with the characteristic function of  $D_p$  on  $M^{|y|}$ . Then  $\text{LS}(\varphi) = S_\varphi(M)$ . Indeed, for  $a \in M$  the left type  $\text{ltp}_\varphi(a)$  is  $\text{tp}_\varphi(a/M)$ . By Lemma 1.1.2, every  $p \in S_\varphi(M)$  agrees on each finite set of parameters with some realized type, so  $p$  is a pointwise limit of realized types. Conversely, a pointwise limit of realized types determines a set of  $\varphi$ -formulas over  $M$  that is finitely satisfiable in  $M$ , hence consistent with the elementary diagram of  $M$ , and so a complete  $\varphi$ -type.

In this two-valued setting, the quotient map  $\beta M^{|x|} \rightarrow S_\varphi(M)$  is transparent. The sets  $\varphi(M, b) = \{a \in M^{|x|} : M \models \varphi(a, b)\}$ , for  $b \in M^{|y|}$ , generate a Boolean algebra  $\mathcal{B}_\varphi(M)$  of subsets of  $M^{|x|}$ ; a complete  $\varphi$ -type is the same thing as an ultrafilter on  $\mathcal{B}_\varphi(M)$ , and  $S_\varphi(M)$  is the Stone space of  $\mathcal{B}_\varphi(M)$ . An ultrafilter  $\mathcal{U}$  on  $M^{|x|}$  goes to its trace on  $\mathcal{B}_\varphi(M)$ , that is, to the type containing  $\varphi(x, b)$  exactly when  $\varphi(M, b) \in \mathcal{U}$ . The realized types are the traces of principal ultrafilters, and Lemma 1.1.2 says that every type is the trace of some ultrafilter.

By Remark 1.2.2, a formula without the order property is stable on every model of  $T$ . So Corollary 1.2.13 gives (1)  $\Rightarrow$  (3) of Theorem 1.1.6. The definitions it produces are Boolean combinations of instances rather than positive ones, and their size depends on  $p$ .



**Remark 1.2.14.** (1) *Lattice version.* The lattice form of the Stone–Weierstrass theorem replaces polynomials by the operations max, min and affine maps. This is the form used in continuous logic, where the lattice form shows that a small set of connectives generates all continuous connectives up to uniform approximation [43, Proposition 1.4].

(2) *Convex combinations.* Stability gives more than a polynomial approximation. In Proposition A.1.6 below we show, using Grothendieck’s criterion, that stability of  $\varphi$  implies that  $\{e_a : a \in A\}$  is relatively weakly compact in  $C(\text{RS}(\varphi))$ . In that case  $p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i)$  implies that  $\hat{\varphi}(p, \cdot)$  is the weak limit of the functions  $e_{a_i}$  along  $\mathcal{U}$ . Mazur’s lemma then shows that  $\hat{\varphi}(p, \cdot)$  is a uniform limit of convex combinations  $\sum_j \lambda_j e_{a_{i_j}}$ . So  $p$  is a uniform limit on  $B$  of averages of instances  $\varphi(a, y)$ . Ben Yaacov [42] uses this convex approximation to recover the familiar forms of the defining formulas.

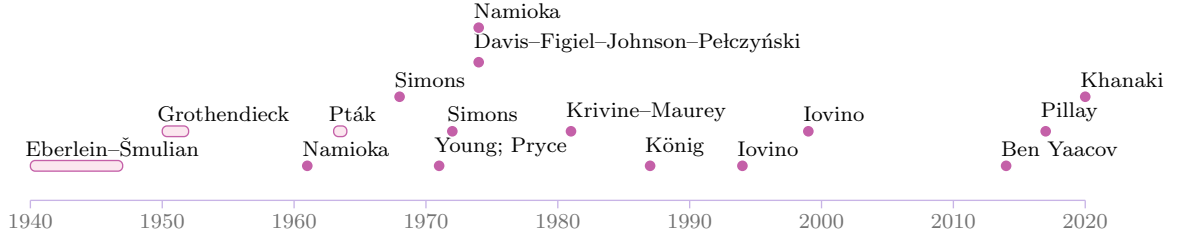
### 1.3. Historical remarks



**Section 1.1.** Counting types begins with Morley’s proof of Loś’s conjecture [35]. There he introduced totally transcendental theories, defined by a restriction on the topology of the type spaces of those theories, and showed that for countable theories total transcendence is equivalent to a restriction on the number of types: only countably many types over countable sets. This is the condition now called  $\omega$ -stability. The general notion arose soon after. Rowbottom set out to extend Morley’s categoricity theorem to uncountable theories, and in the process he generalized  $\omega$ -stability to  $\lambda$ -stability: a theory is  $\lambda$ -stable if there are at most  $\lambda$  types over sets of size  $\lambda$  [66]. Shelah then called a theory *stable* when  $\lambda$ -stable for some  $\lambda$ , and began the study of stable theories in general in [41]. In [36] he moved from theories to single formulas. There he defined when a formula  $\varphi(x; y)$  is *stable in  $\lambda$* : the formula  $\varphi$  has at most  $\lambda$   $\varphi$ -types over sets of size  $\lambda$  [36, Definition 2.1]. He also defined the *order property* [36, Definition 2.2] and proved that stability is equivalent to the absence of the order property. The ladder formulation of Definition 1.1.4 is a finitary restatement of Shelah’s condition. Definable types were introduced by Gaifman in his study of end extensions of models of Peano arithmetic [40], and Shelah arrived at them independently. The theorem that  $\varphi$ -types are definable when  $\varphi$  is stable is [36, Theorem 3.1]; see also [57, Chapter II]. Today this theorem is often called the fundamental theorem of stability theory. Ben Yaacov [42]

observed that the same result follows from Grothendieck’s 1952 work on weak compactness in function spaces [44] (see Appendix A below). This is the viewpoint of Section 1.2.

✂ **Section 1.2.1.** Interchanging iterated limits is an old theme. For double sequences of reals, the classical sufficient condition is uniform convergence in one variable (the Moore–Osgood theorem). The condition of Definition 1.2.5 requires the iterated limits to agree along *all* pairs of sequences. This condition comes from functional analysis, and reached model theory through Banach space geometry.



**1940–1947**

**Eberlein–Šmulian.** Šmulian (1940) and Eberlein [17] (1947) prove that in a Banach space weak compactness and weak sequential compactness coincide, so that weak compactness can be tested along sequences.

**1950–1952**

**Grothendieck.** Grothendieck proves the *double limit criterion*, announced in [18] and proved in [44]. For compact  $K$ , a bounded set  $H \subseteq C(K)$  is relatively weakly compact if and only if  $\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$  for all sequences  $(f_n)$  in  $H$  and  $(x_m)$  in  $K$  for which both iterated limits exist. For an arbitrary topological space, the same condition decides when pointwise limits of functions in  $H$  are continuous. See Appendix A.

**1961**

**Namioka.** Namioka proves a combinatorial “substitute for Lebesgue’s bounded convergence theorem” [23]. From this result he obtains, without measure theory, Krein’s theorem: in a Banach space, the closed convex hull of a weakly compact set is weakly compact.

**1963–1964**

**Pták.** Pták [26] finds a purely combinatorial lemma about averages, from which both the double limit criterion and Krein’s theorem follow. In [27] he uses an interchange of iterated limits to decide which bounded functions on a product  $X \times Y$ , continuous in each variable separately, extend to bilinear forms with the same kind of continuity.

**1968**

**Simons.** Simons [28] introduces the *Fubini characterization* of the interchange of double limits. A bounded function on a product satisfies the iterated limit condition if and only if the two iterated integrals of the function agree, for every pair of measures on the factors. This characterization plays an important role in the proof of the factorization theorem of Davis, Figiel, Johnson and Pełczyński [46].

**1971**

**Young; Pryce.** N. J. Young [32] gives topological proofs of Pták’s double-limit theorems, based on Grothendieck’s work rather than combinatorics, and generalizes Pták’s bilinear extension theorem. Pryce [25] uses a trick



of Whitley's to simplify proofs about pointwise compact sets of continuous functions.

- 1972** • **Simons.** In three consecutive papers, Simons gives an elementary proof of Pták's lemma [31]. He also proves an inequality about suprema and upper limits, now known as *Simons' inequality* [29], and relates minimax theorems, which say when  $\max \min = \min \max$ , to James's characterization of reflexivity [30].
- 1974** • **Davis–Figiel–Johnson–Pełczyński.** Davis, Figiel, Johnson and Pełczyński [46] prove that every weakly compact operator factors through a reflexive Banach space. This is the key step in the proof of Theorem 1.2.9.
- 1974** • **Namioka.** Namioka [24] studies when separately continuous functions on products are jointly continuous at many points. This is the question behind Proposition 1.2.8.
- 1981** • **Krivine–Maurey.** Krivine and Maurey [51] apply the interchange of limits to the norm: a Banach space is *stable* if  $\lim_m \lim_n \|x_m + y_n\| = \lim_n \lim_m \|x_m + y_n\|$  for all bounded sequences for which both limits exist. They prove that every stable Banach space contains, for some  $p$ , copies of  $\ell_p$  that are as close to isometric as desired, which generalizes a theorem of Aldous about subspaces of  $L_1$ .
- 1987** • **König.** König [22] studies the double limit relation in angelic spaces, where compactness can be tested by sequences.
- 1994** • **Iovino.** Iovino's Ph.D. thesis [47] introduces stability for real-valued logic.
- 1999** • **Iovino.** Iovino [49] observes that Shelah's order property is the 2-valued case of the same condition, and proves Theorem 1.2.9. Propositions 1.2.4 and 1.2.3 are taken from his paper.
- 2014** • **Ben Yaacov.** Ben Yaacov [42] derives the definability of types in stable theories directly from Grothendieck's criterion.
- 2017** • **Pillay.** Pillay [21] comments on the model-theoretic reading of Grothendieck's criterion, and derives from the criterion the positive form of the definitions.
- 2020** • **Khanaki.** Khanaki [20] studies stability, NIP (see Chapter 5), and related properties systematically, through topological properties of function spaces.

✂ **Section 1.2.2.** Spaces of left and right  $\varphi$ -types (see Definition 1.2.7) are the analytic counterpart of Shelah's spaces of  $\varphi$ -types. The use of *types* in Banach space theory, as limits of functions  $y \mapsto \|x_n + y\|$ , is due to Krivine and Maurey [51]. Proposition 1.2.8, which characterizes stability by the existence of a separately continuous extension to pairs of types, is Proposition 2.6 of Iovino [49]. Separately continuous functions on products of compact spaces, and their relation to iterated limits, are the subject of Grothendieck's paper [44].

✂ **Section 1.2.3.** James [50] characterized reflexivity by the absence of sequences  $(a_m)$  in  $B_E$  and  $(f_n)$  in  $B_{E^*}$  whose pairing takes one value for  $m < n$  and another for  $m \geq n$ . This is a Banach-space form of the order property. In the early 1980s, Krivine and Maurey [51] and Raynaud [52] showed that stable functions give rise to weakly compact operators. The

factorization theorem of Davis, Figiel, Johnson and Pełczyński [46] (1974) says that every weakly compact operator factors through a reflexive space. Iovino [49] combined these results to prove Theorem 1.2.9: stable formulas are exactly the pairings between the unit balls of a reflexive space  $E$  and of  $E^*$ .

✂ **Section 1.2.4.** Weierstrass proved in 1885 that every continuous function on a compact interval is a uniform limit of polynomials [39]. Stone replaced the interval by an arbitrary compact Hausdorff space and the coordinate function by any family of functions that separates points [37, 38]. Ben Yaacov and Usvyatsov [43] used Stone’s theorem to lay the foundations of continuous first-order logic. There Stone’s theorem shows that finitely many connectives suffice and that continuous functions on type spaces are exactly the definable predicates. Ben Yaacov [42] combined Grothendieck’s criterion with Mazur’s lemma [34] (a weakly convergent sequence has convex combinations converging in norm). From this he derived the defining formulas of stable types as uniform limits of averages of instances; see Remark 1.2.14.



## CHAPTER 2

### Stability, trees, and definability of types

In Chapter 1 we stated the theorem on definability of types (see Theorem 1.1.6) without proof. In this chapter we prove that theorem. The argument passes through binary trees, and along the way we show that a formula is stable exactly when it cannot encode tall trees. As in Section 1.1, we fix a first-order theory  $T$  in a language  $L$  and an  $L$ -formula  $\varphi(x, y)$  consistent with  $T$ .

#### 2.1. The equivalence proved in this chapter

We begin by stating precisely what will be proved. Four conditions on  $\varphi$  turn out to coincide: instability, the order property, the presence of tall binary trees, and the existence of a non-definable  $\varphi$ -type. The notions of  $\varphi$ -type, definable type, ladder, order property and stability were discussed in Section 1.1. The remaining notion, that of a binary tree for  $\varphi$ , is defined below.

**Notation 2.1.1.** Recall that  $\varphi^1 = \varphi$  and  $\varphi^0 = \neg\varphi$ . For an ordinal  $\mu$ , we write  $2^{<\mu}$  for the set of 0–1 sequences of length  $<\mu$ , and we write  $2^{\leq n} = 2^{<n} \cup 2^n$ . We write  $\sigma \sqsubseteq \tau$  if  $\sigma$  is an initial segment of  $\tau$ ,  $\sigma \sqsubset \tau$  if moreover  $\sigma \neq \tau$ ,  $|\sigma|$  for the length of  $\sigma$ ,  $\sigma \frown \langle u \rangle$  for the extension of  $\sigma$  by one entry  $u$ , and, for  $\eta$  of length  $>k$ ,  $\eta \upharpoonright k$  for the initial segment of  $\eta$  of length  $k$ .

**Definition 2.1.2.** A *tree of height  $n$  for  $\varphi$  in a model  $M$  of  $T$*  consists of families  $(b_\sigma : \sigma \in 2^{<n})$  and  $(a_\eta : \eta \in 2^n)$  of tuples of  $M$  such that

$$\sigma \frown \langle u \rangle \sqsubseteq \eta \implies M \models \varphi(a_\eta, b_\sigma)^u.$$

In other words, the leaf  $a_\eta$  satisfies  $\varphi(x, b_\sigma)$  exactly when the branch  $\eta$  turns right ( $u = 1$ ) at the node  $\sigma$ . We say that  $\varphi$  *encodes binary trees of arbitrarily large finite height* if for every  $n$  some model of  $T$  contains a tree of height  $n$  for  $\varphi$ .

Figure 2.1 shows a tree of height 3. Each node  $b_\sigma$  splits the leaves above it into two halves: the leaves on the right satisfy  $\varphi(x, b_\sigma)$ , and the leaves on the left satisfy  $\neg\varphi(x, b_\sigma)$ . So the branch leading to a leaf  $a_\eta$  records, node by node, which of the formulas  $\varphi(x, b_\sigma)$  the leaf satisfies.

A tree of height  $n + 1$  restricts to one of height  $n$ : keep the nodes  $b_\sigma$  with  $\sigma \in 2^{<n}$ , and for each  $\eta \in 2^n$  keep the leaf  $a_{\eta \frown \langle 0 \rangle}$ .

**Theorem 2.1.3.** *Let  $\varphi(x, y)$  be an  $L$ -formula. The following are equivalent.*

- (1) *The formula  $\varphi$  is unstable (see Definition 1.1.4).*
- (2) *The formula  $\varphi$  has the order property (see Definition 1.1.4).*
- (3) *The formula  $\varphi$  encodes binary trees of arbitrarily large finite height.*
- (4) *Some  $\varphi$ -type over some model of  $T$  is not definable (see Definitions 1.1.1 and 1.1.3).*

The proof goes around a cycle of implications. The equivalence (1)  $\Leftrightarrow$  (2) is the definition of stability (see Definition 1.1.4). In Section 2.3 we turn long ladders into tall trees, which gives (2)  $\Rightarrow$  (3) (see Proposition 2.3.1). In Section 2.4 we use tall trees to produce more  $\varphi$ -types than there are formulas to define them, which gives (3)  $\Rightarrow$  (4) (see Proposition 2.4.2).

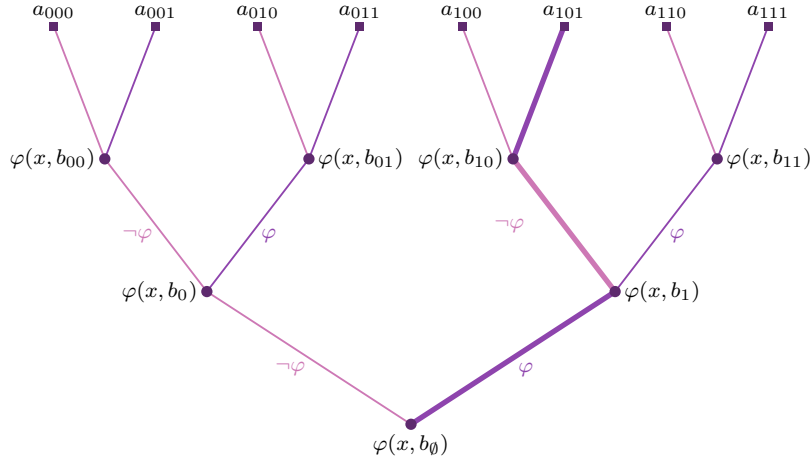


FIGURE 2.1. A tree of height 3 for  $\varphi$ . Each round node is labeled by the formula  $\varphi(x, b_\sigma)$  given by its parameter  $b_\sigma$ , and the squares  $a_\eta$  are the leaves. An edge going up and to the right from the node  $\varphi(x, b_\sigma)$  is labeled  $\varphi$  (purple), and an edge going up and to the left is labeled  $\neg\varphi$  (pink): every leaf above a right edge out of that node satisfies  $\varphi(x, b_\sigma)$ , and every leaf above a left edge satisfies  $\neg\varphi(x, b_\sigma)$ . The edges into the leaves are not labeled, but their colors follow the same rule. For example, the thick branch leading up to  $a_{101}$  goes right, left, right, so  $M \models \varphi(a_{101}, b_\emptyset) \wedge \neg\varphi(a_{101}, b_1) \wedge \varphi(a_{101}, b_{10})$ . The formula  $\varphi$  is not required to say anything about  $a_\eta$  and  $b_\sigma$  when  $\sigma$  is not an initial segment of  $\eta$ , such as  $a_{101}$  and  $b_\emptyset$ .

The cycle closes with the definability theorem, proved in Section 2.2, whose contrapositive is (4)  $\Rightarrow$  (2) (see Theorem 2.2.1 and Corollary 2.2.2).

Chapter 3 gives alternative proofs of two of these implications, and each alternative proof yields information that the proofs in this chapter do not provide. In this chapter, the way from tall trees back to the order property passes through non-definable types, which requires counting types over models of large cardinality. In Section 3.1 we prove (3)  $\Rightarrow$  (2) directly, by finite combinatorics based on Ramsey's theorem (see Proposition 3.1.5 and Corollary 3.1.6). The direct proof stays inside a single model and is quantitative: a tree of height  $2^{\binom{2n}{n}} - 1$  in a model  $M$  already yields an  $n$ -ladder in  $M$ . In Section 3.2, Proposition 3.2.3 proves (4)  $\Rightarrow$  (3), again in contrapositive form, by the classical rank argument. Proposition 3.2.3 shows that the absence of tall trees alone makes every  $\varphi$ -type definable, without going through the order property, and gives the definition explicitly in terms of a rank. Unlike the definitions of Section 2.2, however, these definitions are not positive Boolean combinations of instances of  $\varphi$ .

## 2.2. From the absence of the order property to type definability

In this section we prove (1)  $\Rightarrow$  (2) of Theorem 1.1.6: if  $\varphi$  does not have the order property, then every  $\varphi$ -type over a model has a definition built from instances of  $\varphi$  using only  $\wedge$  and  $\vee$ . The candidate definitions are easy to write down, and the work lies in choosing the finitely many parameters that they use.

Let  $M$  be a model of  $T$ , let  $p \in S_\varphi(M)$  be given, and recall that  $D_p = \{b \in M : \varphi(x, b) \in p\}$ . For a finite set  $F \subseteq M$ , we shall write

$$\mathcal{W}_F = \{\{a \in F : M \models \varphi(a, b)\} : b \in D_p\} \quad \text{and} \quad d_F\varphi(y) = \bigvee_{W \in \mathcal{W}_F} \bigwedge_{a \in W} \varphi(a, y).$$

Since  $\mathcal{W}_F$  is a family of subsets of the finite set  $F$ , the family  $\mathcal{W}_F$  is finite, so  $d_F\varphi$  is a positive Boolean combination of instances of  $\varphi$  over  $M$ . In words, the formula  $d_F\varphi(y)$  says that, for some  $b' \in D_p$ , every  $a \in F$  with  $M \models \varphi(a, b')$  also satisfies  $\varphi(a, y)$ . Every parameter  $b \in D_p$  satisfies  $d_F\varphi$ , whatever  $F$  is (take  $b' = b$ ). The task is to choose  $F$  so that no parameter outside  $D_p$  satisfies  $d_F\varphi$ , and this is where the absence of the order property is used.

**Theorem 2.2.1.** *Assume that  $\varphi(x, y)$  does not have the order property, let  $M$  be a model of  $T$ , and let  $p \in S_\varphi(M)$  be given. Then there is a finite set  $F \subseteq M$  such that  $d_F\varphi$  is a  $\varphi$ -definition of  $p$ ; that is, for all  $b \in M$  we have  $\varphi(x, b) \in p$  if and only if  $M \models d_F\varphi(b)$ .*

*Proof.* For  $b \in M$  we shall write  $S_b = \{a \in M : M \models \varphi(a, b)\}$ . The only property of  $p$  we use is that  $p$  is finitely satisfiable in  $M$  (see Lemma 1.1.2).

*Claim.* There is a finite  $F \subseteq M$  such that  $S_b \cap F \not\subseteq S_{b'}$  for all  $b \in D_p$  and  $b' \in M \setminus D_p$ . Granting the claim, we check that  $d_F\varphi$  defines  $p$ . Let  $b \in M$  be given.

- (a) If  $b \in D_p$ , then  $W := S_b \cap F \in \mathcal{W}_F$  and  $W \subseteq S_b$ , so  $M \models \bigwedge_{a \in W} \varphi(a, b)$ . Hence  $M \models d_F\varphi(b)$ .
- (b) If  $b \notin D_p$  and  $W \in \mathcal{W}_F$ , say  $W = S_{b_0} \cap F$  with  $b_0 \in D_p$ , then  $W \not\subseteq S_b$  by the claim. So some  $a \in W$  has  $M \models \neg\varphi(a, b)$ , and  $M \models \neg\bigwedge_{a \in W} \varphi(a, b)$ . As this holds for every  $W \in \mathcal{W}_F$ , we conclude that  $M \models \neg d_F\varphi(b)$ .

*Proof of the claim.* Suppose the claim fails. Then for every finite  $F \subseteq M$  there are  $b \in D_p$  and  $b' \in M \setminus D_p$  with  $S_b \cap F \subseteq S_{b'}$ , i.e.  $M \models \varphi(a, b) \rightarrow \varphi(a, b')$  for all  $a \in F$ . We choose recursively, for  $i < \omega$ , elements  $b_i \in D_p$ ,  $b'_i \in M \setminus D_p$  and  $a_i \in M$  as follows.

- (a) Apply the failure of the claim to  $F = \{a_j : j < i\}$  to get  $b_i \in D_p$  and  $b'_i \in M \setminus D_p$  with

$$(2.1) \quad M \models \varphi(a_j, b_i) \rightarrow \varphi(a_j, b'_i) \quad \text{for all } j < i.$$

- (b) Since  $\varphi(x, b_k) \in p$  and  $\neg\varphi(x, b'_k) \in p$  for  $k \leq i$ , Lemma 1.1.2 gives  $a_i \in M$  with

$$(2.2) \quad M \models \varphi(a_i, b_k) \wedge \neg\varphi(a_i, b'_k) \quad \text{for all } k \leq i.$$

Color each pair  $j < i$  by the pair of truth values  $(\varphi(a_j, b_i), \varphi(a_j, b'_i)) \in \{0, 1\}^2$ . By (2.1), the color  $(1, 0)$  never occurs. By Ramsey's theorem there is an infinite  $H = \{h_0 < h_1 < \dots\} \subseteq \omega$  on which all pairs have the same color  $c$ . Let  $s, t < \omega$  be given.

*Case  $c \in \{(0, 1), (1, 1)\}$ .* If  $s < t$ , then  $M \models \varphi(a_{h_s}, b'_{h_t})$  by homogeneity. If  $s \geq t$ , then  $M \models \neg\varphi(a_{h_s}, b'_{h_t})$  by (2.2). So  $\varphi(a_{h_s}, b'_{h_t})$  holds if and only if  $s < t$ , and with  $\beta_t = b'_{h_{t+1}}$  we get  $M \models \varphi(a_{h_s}, \beta_t) \iff s \leq t$ .

*Case  $c = (0, 0)$ .* If  $s < t$ , then  $M \models \neg\varphi(a_{h_s}, b_{h_t})$  by homogeneity. If  $s \geq t$ , then  $M \models \varphi(a_{h_s}, b_{h_t})$  by (2.2). So  $\varphi(a_{h_s}, b_{h_t})$  holds if and only if  $s \geq t$ . For a given  $n$ , reversing the first  $n$  indices by putting  $\alpha_s = a_{h_{n-1-s}}$  and  $\beta_t = b_{h_{n-1-t}}$  gives  $M \models \varphi(\alpha_s, \beta_t) \iff s \leq t$  for  $s, t < n$ .

In both cases  $M$  contains an  $n$ -ladder for every  $n$ , so  $\varphi$  has the order property, a contradiction. This proves the claim and the theorem. 

**Corollary 2.2.2.** *If  $\varphi$  does not have the order property, then for every model  $M$  of  $T$ , every  $p \in S_\varphi(M)$  is definable by a positive Boolean combination of instances  $\varphi(a, y)$  with  $a \in M$ . In particular, if  $\varphi$  is quantifier-free, then  $p$  has a quantifier-free  $\varphi$ -definition.*

### 2.3. From the order property to tall trees

To prove (3)  $\Rightarrow$  (1) of Theorem 1.1.6, we need a supply of many  $\varphi$ -types over a small model. Binary trees (see Definition 2.1.2) provide such a supply, and in this section we show that long ladders produce tall trees. The idea is to place the nodes and the leaves of a binary tree on a line, in such a way that the leaves above the right edge out of each node lie to the left of that node and the leaves above the left edge lie to the right of the node. A ladder read along this line then has exactly the pattern of a tree.

**Proposition 2.3.1.** *If a model  $M$  of  $T$  contains an  $N$ -ladder for  $\varphi$ , where  $N = 2^{n+1} - 1$ , then  $M$  contains a tree of height  $n$ . Hence, if  $\varphi$  has the order property, then  $\varphi$  encodes binary trees of arbitrarily large finite height.*

*Proof.* For  $\sigma \in 2^{\leq n}$  put

$$d(\sigma) = \sum_{k < |\sigma|} (1 - \sigma(k)) 2^{-(k+1)} + 2^{-(|\sigma|+1)} \in (0, 1).$$

The binary expansion of  $d(\sigma)$  is  $(1 - \sigma(0), \dots, 1 - \sigma(|\sigma| - 1), 1)$ , which ends in 1. So  $d$  is injective on  $2^{\leq n}$ , and the image  $V$  of  $d$  has exactly  $|2^{\leq n}| = 2^{n+1} - 1 = N$  elements.

*Claim.* If  $\sigma \frown \langle u \rangle \sqsubseteq \eta$  with  $\eta \in 2^n$ , then  $d(\eta) < d(\sigma) \iff u = 1$ .

Indeed, let  $k = |\sigma|$ . Since  $\eta|_k = \sigma$  and  $\eta(k) = u$ , we have

$$d(\eta) - d(\sigma) = -u 2^{-(k+1)} + R, \quad R = \sum_{k < j < n} (1 - \eta(j)) 2^{-(j+1)} + 2^{-(n+1)}.$$

Here  $0 < R \leq (2^{-(k+1)} - 2^{-n}) + 2^{-(n+1)} < 2^{-(k+1)}$ . Hence  $d(\eta) - d(\sigma)$  is negative if  $u = 1$  and positive if  $u = 0$ , which proves the claim.

Let  $a_1, \dots, a_N, b_1, \dots, b_N \in M$  satisfy  $\varphi(a_i, b_j) \iff i \leq j$ , and let  $e: V \rightarrow \{1, \dots, N\}$  be the order isomorphism. Put  $b_\sigma := b_{e(d(\sigma))}$  for  $\sigma \in 2^{\leq n}$  and  $a_\eta := a_{e(d(\eta))}$  for  $\eta \in 2^n$ . If  $\sigma \frown \langle u \rangle \sqsubseteq \eta$ , then  $d(\eta) \neq d(\sigma)$ , and so

$$M \models \varphi(a_\eta, b_\sigma) \iff e(d(\eta)) \leq e(d(\sigma)) \iff d(\eta) < d(\sigma) \iff u = 1.$$

Thus  $(b_\sigma), (a_\eta)$  is a tree of height  $n$  in  $M$ . ☺

### 2.4. Tall trees produce non-definable types

In this section we prove (3)  $\Rightarrow$  (4) of Theorem 2.1.3: tall trees produce more  $\varphi$ -types over a small model than there are formulas to define them (see Proposition 2.4.2 below). We first use compactness to turn finite trees into a tree of infinite height (see Lemma 2.4.1 below).

**Lemma 2.4.1 (compactness).** *Suppose that  $\varphi$  encodes binary trees of arbitrarily large finite height, and let  $\mu$  be an infinite cardinal. Then there are a model  $M'$  of  $T$  and tuples  $(b_s)_{s \in 2^{< \mu}}$  in  $M'$  such that for every  $\eta \in 2^\mu$  the set*

$$\Sigma_\eta(x) = \{\varphi(x, b_{\eta|_\alpha})^{\eta(\alpha)} : \alpha < \mu\}$$

*is finitely satisfiable in  $M'$ .*

*Proof.* Add new tuples of constants  $c_s$  ( $s \in 2^{<\mu}$ ) and let  $\Gamma$  be  $T$  together with the sentences

$$\gamma_{\eta,F} := \exists x \bigwedge_{\alpha \in F} \varphi(x, c_{\eta|\alpha})^{\eta(\alpha)} \quad (\eta \in 2^\mu, F \subseteq \mu \text{ finite}).$$

By compactness, we need only show that every finite  $\Gamma_0 \subseteq \Gamma$  has a model.

Let  $\gamma_{\eta_i, F_i}$  ( $i < m$ ) be the sentences of this form in  $\Gamma_0$  (if  $m = 0$  there is nothing to do). Let  $S$  be the finite set of all  $\eta_i|\alpha$  and  $\eta_i|(\alpha + 1)$  with  $i < m$  and  $\alpha \in F_i$ . Note that  $\alpha + 1 < \mu$  because  $\mu$  is a limit ordinal. Let  $S' = \{s \wedge s' : s, s' \in S\}$ , where  $s \wedge s'$  is the longest common initial segment. The set  $S'$  is finite and closed under  $\wedge$ , because in a tree  $s \wedge s' \wedge s''$  equals one of the pairwise meets. So  $S'$  has a least element  $r$ , and every  $s \in S' \setminus \{r\}$  has a longest predecessor  $s^- \in S'$  with  $s^- \sqsubset s$ .

Define  $f: S' \rightarrow 2^{<N}$ , where  $N = |S'| + 1$ , by recursion on length, setting  $f(r) = \emptyset$  and  $f(s) = f(s^-) \frown \langle s(|s^-|) \rangle$ . Then  $|f(s)| < N$ .

*Claim.* If  $s, s' \in S'$  and  $s \sqsubset s'$ , then  $f(s) \frown \langle s'(|s|) \rangle \sqsubseteq f(s')$ .

The proof is by induction on  $|s'|$ . If  $s'^- = s$ , this is the definition of  $f$ . Otherwise  $s \sqsubset s'^- \sqsubset s'$  by maximality of  $s'^-$ . By induction, we have  $f(s) \frown \langle s'^- (|s|) \rangle \sqsubseteq f(s'^-) \sqsubseteq f(s')$ , and finally  $s'^- (|s|) = s'(|s|)$ .

By hypothesis, take a model  $M_0$  of  $T$  containing a tree  $(b_\sigma)_{\sigma \in 2^{<N}}$ ,  $(a_\nu)_{\nu \in 2^N}$  of height  $N$ . Interpret  $c_s$  as  $b_{f(s)}$  for  $s \in S'$ , and arbitrarily otherwise. Fix  $i < m$ , let  $\alpha^* = \max F_i$  and  $s^* = \eta_i|(\alpha^* + 1) \in S'$ . Pick  $\nu \in 2^N$  extending  $f(s^*)$ . For  $\alpha \in F_i$  we have  $\eta_i|\alpha \sqsubset s^*$  and  $s^*(\alpha) = \eta_i(\alpha)$ . By the Claim, we have  $f(\eta_i|\alpha) \frown \langle \eta_i(\alpha) \rangle \sqsubseteq \nu$ , and so  $M_0 \models \varphi(a_\nu, b_{f(\eta_i|\alpha)})^{\eta_i(\alpha)}$ . Hence  $a_\nu$  witnesses  $\gamma_{\eta_i, F_i}$ , and  $M_0$  with this interpretation is a model of  $\Gamma_0$ .

By compactness  $\Gamma$  has a model. The  $L$ -reduct  $M'$  of that model, with  $b_s = c_s^{M'}$ , is as required.  $\odot$

In the next proposition we write  $\kappa = |L| + \aleph_0$ , which is the number of  $L$ -formulas.

**Proposition 2.4.2.** *If  $\varphi$  encodes binary trees of arbitrarily large finite height, then for every cardinal  $\lambda \geq \kappa$  there is a model  $M$  of  $T$  with  $|M| \leq \lambda$  and  $|S_\varphi(M)| > \lambda$ . In particular, there are a model  $M$  of  $T$  with  $|M| \leq \kappa$  and a type  $p \in S_\varphi(M)$  such that  $p$  is not definable.*

*Proof.* Let  $\mu$  be the least cardinal with  $2^\mu > \lambda$ . Then  $\mu$  is infinite and  $\mu \leq \lambda$ . For  $\nu < \mu$  we have  $2^\nu \leq \lambda$ , so  $|2^{<\mu}| = \sum_{\nu < \mu} 2^\nu \leq \mu \cdot \lambda = \lambda$ .

Take  $M'$  and  $(b_s)$  as in Lemma 2.4.1. By the downward Löwenheim–Skolem theorem, there is  $M \preceq M'$  with  $|M| \leq \lambda$  containing all  $b_s$ . Each finite part of  $\Sigma_\eta$  is satisfiable in  $M'$ , hence in  $M$ , since  $\exists x \bigwedge \dots$  is an  $L(M)$ -sentence. So  $\Sigma_\eta$  is consistent with the elementary diagram of  $M$  and extends to some  $p_\eta \in S_\varphi(M)$ .

If  $\eta \neq \eta'$ , let  $\alpha$  be least with  $\eta(\alpha) \neq \eta'(\alpha)$ . Then  $s = \eta|\alpha = \eta'|\alpha$ , and  $p_\eta$  and  $p_{\eta'}$  disagree on  $\varphi(x, b_s)$ . Hence  $|S_\varphi(M)| \geq 2^\mu > \lambda$ .

For the last statement take  $\lambda = \kappa$ . A definable type is determined by any  $\varphi$ -definition of the type, and there are only  $|L(M)| \leq \kappa$  formulas over  $M$ . So at most  $\kappa$  types in  $S_\varphi(M)$  are definable, and some  $p \in S_\varphi(M)$  is not.  $\odot$

*Proof of (3)  $\Rightarrow$  (1) in Theorem 1.1.6.* If  $\varphi$  has the order property, then  $\varphi$  encodes binary trees of arbitrarily large finite height by Proposition 2.3.1, and so some  $\varphi$ -type over some model of  $T$  is not definable by Proposition 2.4.2.  $\odot$

This completes the proof of Theorem 1.1.6. We record the tree characterization that came out of this proof.

**Theorem 2.4.3.** *The formula  $\varphi$  has the order property if and only if  $\varphi$  encodes binary trees of arbitrarily large finite height.*

*Proof.* The forward direction is Proposition 2.3.1. Conversely, if  $\varphi$  encodes tall trees, then some  $\varphi$ -type is not definable (see Proposition 2.4.2), so  $\varphi$  has the order property by (1)  $\Rightarrow$  (3) of Theorem 1.1.6.  $\odot$

## 2.5. Type spaces over countable models: a perfect set dichotomy

Section 2.4 measured instability by counting types; topology measures instability by shape. In this section the language  $L$  is countable. For a countable model  $M$  of  $T$ , the Boolean algebra  $\mathcal{B}_\varphi(M)$  of Subsection 1.2.4 is countable, so its Stone space  $S_\varphi(M)$  is a compact metrizable zero-dimensional space. By the Cantor–Bendixson theorem, such a space is either countable, in which case the space is scattered, or contains a homeomorphic copy of the Cantor space  $2^\omega$ . Stability decides which alternative occurs.

**Theorem 2.5.1.** *Assume that  $L$  is countable. The following are equivalent.*

- (1) *The formula  $\varphi$  is stable.*
- (2) *For every countable model  $M$  of  $T$ , the space  $S_\varphi(M)$  is countable.*
- (3) *For every countable model  $M$  of  $T$ , the space  $S_\varphi(M)$  is scattered.*

*Moreover, if  $\varphi$  is unstable, then some countable model  $M$  of  $T$  has a closed subset of  $S_\varphi(M)$  homeomorphic to  $2^\omega$ .*

*Proof.* (1)  $\Rightarrow$  (2). By Theorem 1.1.6, every  $p \in S_\varphi(M)$  has a  $\varphi$ -definition, which is an  $L(M)$ -formula, and the definition determines  $p$ . There are only countably many  $L(M)$ -formulas.

(2)  $\Leftrightarrow$  (3). This is the Cantor–Bendixson theorem for compact metrizable spaces.

(2)  $\Rightarrow$  (1) and the last statement. Suppose that  $\varphi$  is unstable. By Theorem 2.4.3, the formula  $\varphi$  encodes binary trees of arbitrarily large finite height, so Lemma 2.4.1 with  $\mu = \omega$  gives a model  $M'$  of  $T$  and tuples  $(b_\sigma)_{\sigma \in 2^{<\omega}}$  in  $M'$  such that every set  $\Sigma_\eta(x)$ , for  $\eta \in 2^\omega$ , is finitely satisfiable in  $M'$ . By the downward Löwenheim–Skolem theorem, there is a countable model  $M \preceq M'$  containing every  $b_\sigma$ , and by elementarity every  $\Sigma_\eta(x)$  is finitely satisfiable in  $M$ . For  $\sigma \in 2^{<\omega}$ , let  $C_\sigma$  be the clopen set of types  $p \in S_\varphi(M)$  that contain  $\varphi(x, b_{\sigma|\alpha})^{\sigma(\alpha)}$  for every  $\alpha < |\sigma|$ . Each  $C_\sigma$  is nonempty, because the formulas in question form a finitely satisfiable subset of  $\Sigma_\eta$  for any  $\eta$  extending  $\sigma$ , and  $C_\sigma$  is the disjoint union of  $C_{\sigma \smallfrown \langle 0 \rangle}$  and  $C_{\sigma \smallfrown \langle 1 \rangle}$ , which disagree on  $\varphi(x, b_\sigma)$ . So the sets  $C_\sigma$  form a Cantor scheme. The closed set  $D = \bigcap_n \bigcup_{|\sigma|=n} C_\sigma$  maps continuously onto  $2^\omega$ , by sending  $p$  to the unique  $\eta$  with  $p \in C_{\eta|n}$  for every  $n$ ; the map is onto because each  $\bigcap_n C_{\eta|n}$  is a decreasing intersection of nonempty closed sets in a compact space. Hence  $S_\varphi(M)$  is uncountable, so (2) fails, and by the Cantor–Bendixson theorem the uncountable compact metrizable space  $S_\varphi(M)$  contains a closed copy of  $2^\omega$ .  $\odot$

A tree of infinite height is thus nothing but a Cantor scheme of clopen sets in a type space, and stability says that type spaces over countable models are scattered. This is the same topology that Morley used to define totally transcendental theories, whose type spaces over countable sets are scattered, and whose Morley rank is the Cantor–Bendixson rank on type spaces over  $\aleph_0$ -saturated models. The rank  $R$  of Section 3.2 is a binary splitting rank on the clopen sets  $[\theta]$ , a finite combinatorial cousin of the Cantor–Bendixson rank: Lemma 3.2.2(iii) says that splitting a clopen set into two pieces of rank at least  $m$  raises the rank.

## 2.6. Historical remarks

$\boxtimes$  **Section 2.2.** Definability of  $\varphi$ -types over models for formulas without the order property is due to Shelah [36, 57]. The sharper form proved here, in which the definition is a positive Boolean combination of instances of  $\varphi$  itself, goes back to Hrushovski and Pillay [19], as Pillay

points out in [56, Chapter 1]. Pillay [21, Remark 2.4] derives the positive form directly from Grothendieck's double limit criterion (see Theorem A.1.3 below) by a compactness argument. Positivity matters in applications: for example, positivity gives quantifier-free definitions for quantifier-free formulas.

⌘ **Section 2.3.** Binary trees enter stability theory through Shelah's local rank  $R(-, \varphi, 2)$ . In [36, Lemma 2.7] the rank of a type is computed by the consistency of the family of formulas indexed by the branches of a binary tree of height  $n$ . That is, the formulas of infinite 2-rank are exactly those that encode binary trees of arbitrarily large finite height (see Definition 2.1.2). Shelah showed that this happens exactly when  $\varphi$  has the order property [36, 57]. The two directions behave very differently quantitatively. A ladder of length exponential in  $n$  yields a tree of height  $n$  (see Proposition 2.3.1). In the other direction, a tree yields only a ladder of length logarithmic in the height of the tree; see Section 3.1.

⌘ **Section 2.4.** The construction of this section is the local form of Shelah's counting argument [36]. A tree of infinite height gives  $2^{\aleph_0}$   $\varphi$ -types over a countable set of parameters, which is incompatible with definability of all  $\varphi$ -types. This is also the origin of the *stability spectrum*: a theory with the order property has  $2^\lambda$  types over some set of size  $\lambda$ , for every infinite  $\lambda$  [57].





## CHAPTER 3

### More stability, trees, and definability

This chapter is a complement to Chapter 2. The proofs given there take a short route, at a price: the way from tall trees back to the order property goes through non-definable types and a count of types over large models, and so says nothing about how tall a tree must be to contain a ladder of a given length. In Section 3.1 we answer that question with a direct combinatorial argument based on Ramsey's theorem. In Section 3.2 we present the classical proof of definability of types, which uses ranks instead of the order property. The conventions are those of Chapter 2.

#### 3.1. A quantitative combinatorial proof (using Ramsey's theorem) that trees give the order property

Theorem 2.4.3 was derived from the definability theorem. Here we give a direct and quantitative proof of the implication “tall trees  $\Rightarrow$  order property”. The proof is finite combinatorics: the Boolean case of a Ramsey-type argument due to Hodges. We follow the presentation of Conant and Terry [45].

**Definition 3.1.1.** Let  $s, t \geq 1$  be given.

- (a) A *node embedding* is a map  $\theta: 2^{<s} \rightarrow 2^{<t}$  such that  $\rho \frown \langle u \rangle \sqsubseteq \tau$  implies  $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\tau)$ .
- (b) A *full embedding* is a map  $\theta: 2^{\leq s} \rightarrow 2^{\leq t}$  sending  $2^{<s}$  into  $2^{<t}$  and  $2^s$  into  $2^t$ , with the same property for all  $\rho, \tau \in 2^{\leq s}$ .

**Lemma 3.1.2.** *Every node embedding  $\theta: 2^{<s} \rightarrow 2^{<t}$  extends to a full embedding. If  $(b_\sigma)$ ,  $(a_\eta)$  is a tree of height  $t$  in  $M$  and  $\theta$  is a full embedding  $2^{\leq s} \rightarrow 2^{\leq t}$ , then  $(b_{\theta(\sigma)} : \sigma \in 2^{<s})$ ,  $(a_{\theta(\eta)} : \eta \in 2^s)$  is a tree of height  $s$  in  $M$ .*

*Proof.* For  $\eta \in 2^s$ , let  $\theta(\eta)$  be any element of  $2^t$  extending  $\theta(\eta|(s-1)) \frown \langle \eta(s-1) \rangle$ . Suppose that  $\rho \frown \langle u \rangle \sqsubseteq \eta$ . If  $\rho = \eta|(s-1)$ , the requirement holds by construction. Otherwise  $\rho \frown \langle u \rangle \sqsubseteq \eta|(s-1)$ , so  $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\eta|(s-1)) \sqsubseteq \theta(\eta)$ . The second statement is immediate, since whether  $\varphi(a_\eta, b_\sigma)$  holds depends only on the direction  $u$  with  $\sigma \frown \langle u \rangle \sqsubseteq \eta$ , and full embeddings preserve this direction. ☺

In the next proof and in the rest of this chapter, for  $u \in \{0, 1\}$  and a finite 0–1 sequence  $\sigma$ , we write  $\langle u \rangle \frown \sigma$  for the sequence obtained by putting  $u$  in front of  $\sigma$ .

**Lemma 3.1.3 (Ramsey lemma for trees).** *Let  $t_0, t_1 \geq 1$  be given, let  $t = t_0 + t_1 - 1$ , and suppose that  $2^{<t} = C_0 \cup C_1$ . Then for some  $u \in \{0, 1\}$  there is a node embedding  $\theta: 2^{<t_u} \rightarrow 2^{<t}$  with image contained in  $C_u$ .*

*Proof.* Induction on  $t$ . If  $t = 1$ , then  $t_0 = t_1 = 1$ , and  $\emptyset \mapsto \emptyset$  works for the  $u$  with  $\emptyset \in C_u$ . Now suppose that  $t > 1$ , and choose  $w$  with  $\emptyset \in C_w$ . If  $t_w = 1$ , the map  $\emptyset \mapsto \emptyset$  works, so assume that  $t_w \geq 2$ . For  $v \in \{0, 1\}$  and  $i \in \{0, 1\}$  let  $C_i^v = \{\sigma \in 2^{<t-1} : \langle v \rangle \frown \sigma \in C_i\}$ . Then  $C_0^v \cup C_1^v$  covers  $2^{<t-1}$ , and  $(t_w - 1) + t_{1-w} - 1 = t - 1$ . By induction, for each  $v$  either

- (i) there is a node embedding  $\chi: 2^{<t_1-w} \rightarrow C_{1-w}^v$ , in which case  $\sigma \mapsto \langle v \rangle \frown \chi(\sigma)$  is a node embedding  $2^{<t_1-w} \rightarrow 2^{<t}$  with image in  $C_{1-w}$ , and we are done; or
- (ii) there is a node embedding  $\chi_v: 2^{<t_w-1} \rightarrow C_w^v$ .

If (ii) holds for both  $v$ , define  $\theta: 2^{<t_w} \rightarrow 2^{<t}$  by  $\theta(\emptyset) = \emptyset$  and  $\theta(\langle v \rangle \frown \sigma) = \langle v \rangle \frown \chi_v(\sigma)$ . The image of  $\theta$  lies in  $C_w$ . Suppose that  $\rho \frown \langle u \rangle \sqsubseteq \tau$ . If  $\rho = \emptyset$  then  $\tau = \langle u \rangle \frown \sigma$  and  $\emptyset \frown \langle u \rangle \sqsubseteq \theta(\tau)$ . If  $\rho = \langle v \rangle \frown \rho'$  then  $\tau = \langle v \rangle \frown \tau'$  with  $\rho' \frown \langle u \rangle \sqsubseteq \tau'$ , so  $\chi_v(\rho') \frown \langle u \rangle \sqsubseteq \chi_v(\tau')$ . In both cases  $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\tau)$ , so  $\theta$  is a node embedding.  $\text{☺}$

**Definition 3.1.4.** Let  $(b_\sigma)_{\sigma \in 2^{<t}}$ ,  $(a_\eta)_{\eta \in 2^t}$  be a tree in  $M$ . An *up-ladder of length  $k$*  in this tree consists of nodes  $\sigma_1, \dots, \sigma_k$  and leaves  $\eta_1, \dots, \eta_k$  such that  $M \models \varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$ . A *down-ladder of length  $\ell$*  is defined in the same way with “ $p > q$ ” in place of “ $p \leq q$ ”.

Define  $B(k, \ell)$  for  $k, \ell \geq 1$  by  $B(k, 1) = B(1, \ell) = 1$  and  $B(k, \ell) = B(k-1, \ell) + B(k, \ell-1) + 1$  for  $k, \ell \geq 2$ . By induction on  $k + \ell$ , using Pascal’s rule  $\binom{k+\ell-3}{k-2} + \binom{k+\ell-3}{k-1} = \binom{k+\ell-2}{k-1}$ , we have  $B(k, \ell) = 2^{\binom{k+\ell-2}{k-1}} - 1$ . The recursion is the Erdős–Szekeres recursion for the Ramsey numbers  $R(k, \ell)$ , and the bound mirrors  $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$ ; the proof below follows the Erdős–Szekeres pattern as well. We fix a leaf  $\eta_*$  and split the nodes according to whether  $a_{\eta_*}$  satisfies the corresponding instance of  $\varphi$ . Lemma 3.1.3 finds a large subtree whose nodes all lie on one side of the split, induction finds a ladder inside that subtree, and the pair formed by  $\eta_*$  and the root of the subtree extends the ladder by one rung.

**Proposition 3.1.5.** *For all  $k, \ell \geq 1$ , every tree of height  $B(k, \ell)$  for  $\varphi$  in a model of  $T$  contains an up-ladder of length  $k$  or a down-ladder of length  $\ell$ .*

*Proof.* Induction on  $k + \ell$ . In a tree of height 1, the root  $\emptyset$  with the leaf  $\langle 1 \rangle$  is an up-ladder of length 1, and the root with the leaf  $\langle 0 \rangle$  is a down-ladder of length 1. This settles the case  $k = 1$  or  $\ell = 1$ .

Let  $k, \ell \geq 2$  be given, let  $t = B(k, \ell)$ , and let  $(b_\sigma), (a_\eta)$  be a tree of height  $t$  in  $M$ . Fix a leaf  $\eta_* \in 2^t$  and put

$$P^+ = \{\sigma \in 2^{<t} : M \models \varphi(a_{\eta_*}, b_\sigma)\}, \quad P^- = 2^{<t} \setminus P^+.$$

Let  $t^+ = B(k-1, \ell) + 1$  and  $t^- = B(k, \ell-1) + 1$ , so that  $t^+ + t^- - 1 = t$ . By Lemma 3.1.3, one of the two cases below occurs.

*Case +:* there is a node embedding  $\theta: 2^{<t^+} \rightarrow 2^{<t}$  with image in  $P^+$ . Extend  $\theta$  to a full embedding  $\hat{\theta}$  (see Lemma 3.1.2) and set  $\psi(\tau) = \hat{\theta}(\langle 0 \rangle \frown \tau)$  for  $\tau \in 2^{\leq t^+-1}$ . Then  $\psi$  is a full embedding of height  $B(k-1, \ell)$ . Put  $\sigma_0 = \theta(\emptyset)$ . Since  $\emptyset \frown \langle 0 \rangle \sqsubseteq \langle 0 \rangle \frown \tau$ , we have  $\sigma_0 \frown \langle 0 \rangle \sqsubseteq \psi(\tau)$  for every  $\tau$ . By Lemma 3.1.2, the embedding  $\psi$  induces a tree of height  $B(k-1, \ell)$  in  $M$ . By induction, the induced tree contains a down-ladder of length  $\ell$ , which is then a down-ladder in the original tree, or an up-ladder  $(\sigma'_p, \eta'_p)_{p=1}^{k-1}$ . In the latter case, let

$$(\sigma_1, \eta_1) = (\sigma_0, \eta_*), \quad (\sigma_{p+1}, \eta_{p+1}) = (\psi(\sigma'_p), \psi(\eta'_p)) \quad (1 \leq p < k).$$

We check that  $\varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$ .

- (a) For  $p = 1$ , every  $\sigma_q$  is in the image of  $\theta$ , which lies in  $P^+$ , so  $\varphi(a_{\eta_*}, b_{\sigma_q})$  holds, and indeed  $1 \leq q$ .
- (b) For  $p \geq 2$  and  $q = 1$ , we have  $\sigma_0 \frown \langle 0 \rangle \sqsubseteq \eta_p$ , so  $\neg \varphi(a_{\eta_p}, b_{\sigma_0})$ , and indeed  $p > 1$ .
- (c) For  $p, q \geq 2$ : this is the up-ladder property of the smaller tree.

This is an up-ladder of length  $k$ .

*Case -:* there is a node embedding  $\theta: 2^{<t^-} \rightarrow 2^{<t}$  with image in  $P^-$ . Proceed symmetrically, with  $\psi(\tau) = \hat{\theta}(\langle 1 \rangle \frown \tau)$ , a full embedding of height  $B(k, \ell-1)$  whose leaves all extend  $\sigma_0 \frown \langle 1 \rangle$ .

By induction the induced tree contains an up-ladder of length  $k$ , and we are done, or a down-ladder  $(\sigma'_p, \eta'_p)_{p=1}^{\ell-1}$ . Prepend  $(\sigma_0, \eta_*)$  as before.

- (a) For  $p = 1$ , the formula  $\neg\varphi(a_{\eta_*}, b_{\sigma_q})$  holds for all  $q$ , because the nodes lie in  $P^-$ , and indeed  $1 \not\geq q$ .
- (b) For  $p \geq 2$  and  $q = 1$ , the formula  $\varphi(a_{\eta_p}, b_{\sigma_0})$  holds, because  $\sigma_0 \frown \langle 1 \rangle \sqsubseteq \eta_p$ , and indeed  $p > 1$ .

This is a down-ladder of length  $\ell$ . ☺

**Corollary 3.1.6.** *If a model  $M$  of  $T$  contains a tree of height  $B(n+1, n+1)$  for  $\varphi$ , then  $M$  contains an  $n$ -ladder for  $\varphi$ .*

*Proof.* By Proposition 3.1.5, the model  $M$  contains an up-ladder or a down-ladder of length  $n+1$ . An up-ladder of length  $n+1$  is an  $(n+1)$ -ladder, and so contains an  $n$ -ladder. A down-ladder of length  $n+1$  yields an  $n$ -ladder by Remark 1.1.5. ☺

### 3.2. Optional: The classical proof of definability, via ranks

In this section we give the classical argument that, in the absence of tall trees, every  $\varphi$ -type is definable (see Proposition 3.2.3 below). Together with Proposition 2.3.1, this argument gives another route to (1)  $\Rightarrow$  (3) in Theorem 1.1.6. The definitions obtained in this section, however, are not positive Boolean combinations of instances of  $\varphi$ : they contain quantifiers and negations that do not occur in  $\varphi$  (see Lemma 3.2.2(iv)). Assume that there is  $n$  such that no model of  $T$  contains a tree of height  $n$ , and fix a model  $M$  of  $T$ . By a  $\varphi$ -conjunction over  $M$  we mean a finite conjunction  $\theta(x)$  of formulas  $\varphi(x, b)^u$  with  $b \in M$ .

**Definition 3.2.1.** For a  $\varphi$ -conjunction  $\theta$  over  $M$  and  $m < \omega$ , we write  $R(\theta) \geq m$  if there are  $b_\sigma \in M$  ( $\sigma \in 2^{<m}$ ) such that for every  $\eta \in 2^m$ , we have

$$M \models \exists x \left( \theta(x) \wedge \bigwedge_{k < m} \varphi(x, b_{\eta|k})^{\eta(k)} \right).$$

Thus  $R(\theta) \geq 0$  if and only if  $M \models \exists x \theta(x)$ . When  $\theta$  is satisfiable in  $M$ , we write  $R(\theta)$  for the largest  $m$  with  $R(\theta) \geq m$ .

In Lemma 3.2.2(i) below we show that this largest  $m$  exists.

**Lemma 3.2.2.**

- (i) *The inequality  $R(\theta) < n$  holds for every  $\theta$ ; in particular  $R(\theta)$  is well defined whenever  $\theta$  is satisfiable in  $M$ .*
- (ii) *If  $\theta' = \theta \wedge \chi$ , then  $R(\theta') \leq R(\theta)$ .*
- (iii) *If  $b \in M$  and  $R(\theta \wedge \varphi(x, b)^u) \geq m$  for both  $u \in \{0, 1\}$ , then  $R(\theta) \geq m+1$ .*
- (iv) *For each  $\theta$  and  $m$  there is an  $L(M)$ -formula  $\psi_{\theta, m}(y)$  such that for all  $b \in M$  we have  $R(\theta \wedge \varphi(x, b)) \geq m \iff M \models \psi_{\theta, m}(b)$ .*

*Proof.* (i) If  $R(\theta) \geq n$ , witnessed by  $(b_\sigma)$ , then choosing a witness  $a_\eta \in M$  for each  $\eta \in 2^n$  gives a tree of height  $n$  in  $M$ , which is impossible.

(ii) A witness for  $R(\theta') \geq m$  is a witness for  $R(\theta) \geq m$ .

(iii) Let  $(b_\sigma^u)_{\sigma \in 2^{<m}}$  witness  $R(\theta \wedge \varphi(x, b)^u) \geq m$ . Put  $b_\emptyset = b$  and  $b_{\langle u \rangle \frown \sigma} = b_\sigma^u$ . A branch of length  $m+1$  has the form  $\langle u \rangle \frown \eta$  with  $\eta \in 2^m$ , and the corresponding formula is  $\theta \wedge \varphi(x, b)^u \wedge \bigwedge_{k < m} \varphi(x, b_{\eta|k}^u)^{\eta(k)}$ , which is satisfiable in  $M$ .

(iv) Take

$$\psi_{\theta, m}(y) := \exists (z_\sigma)_{\sigma \in 2^{<m}} \bigwedge_{\eta \in 2^m} \exists x \left( \theta(x) \wedge \varphi(x, y) \wedge \bigwedge_{k < m} \varphi(x, z_{\eta|k})^{\eta(k)} \right),$$

whose parameters are those of  $\theta$ . ☺

**Proposition 3.2.3.** *Let  $M$  be a model of  $T$ . If for some  $n$  no model of  $T$  contains a tree of height  $n$ , then every  $p \in S_\varphi(M)$  is definable.*

*Proof.* Let  $\Phi_p$  be the set of finite conjunctions of members of  $p$ . Each  $\theta \in \Phi_p$  is satisfiable in  $M$ , because the sentence  $\exists x \theta$  is an  $L(M)$ -sentence consistent with the elementary diagram of  $M$ , hence true in  $M$ . By Lemma 3.2.2(i), choose  $\theta \in \Phi_p$  with  $m = R(\theta)$  minimal. We show that  $\psi_{\theta,m}$  is a  $\varphi$ -definition of  $p$ , i.e. that for  $b \in M$ , we have

$$\varphi(x, b) \in p \iff R(\theta \wedge \varphi(x, b)) \geq m.$$

If  $\varphi(x, b) \in p$ , then  $\theta \wedge \varphi(x, b) \in \Phi_p$ , so  $R(\theta \wedge \varphi(x, b)) \geq m$  by minimality. Conversely, suppose that  $\varphi(x, b) \notin p$ . Then  $\neg\varphi(x, b) \in p$ , so  $R(\theta \wedge \neg\varphi(x, b)) \geq m$  by minimality. If also  $R(\theta \wedge \varphi(x, b)) \geq m$ , then Lemma 3.2.2(iii) would give  $R(\theta) \geq m + 1$ , a contradiction. The equivalence together with Lemma 3.2.2(iv) shows that  $p$  is definable. ☺



**Remark 3.2.4.**

- (a) The arguments are effective. An up- or down-ladder of length  $n + 1$  appears in any tree of height  $B(n + 1, n + 1) = 2^{\binom{2n}{n}} - 1$ . Conversely, a ladder of length  $2^{n+1} - 1$  yields a tree of height  $n$ .
- (b) Theorem 2.2.1 uses only that  $p$  is finitely satisfiable in  $M$  (see Lemma 1.1.2). So the theorem applies to any  $\varphi$ -type over a set  $A$  that is finitely satisfiable in  $A$ . The parameters of that definition come from the finite set  $F$ , whereas the rank argument of Proposition 3.2.3 yields a definition whose parameters are those of the formula  $\theta$  of minimal rank.
- (c) Without finite satisfiability, definitions over the parameter set cannot be expected in general, and the restriction to types over models is essential. Definition 3.2.5 and Example 3.2.6 below make this precise.

**Definition 3.2.5.** Let  $N$  be a model of  $T$ , and let  $A \subseteq N$  and  $p \in S_\varphi(N)$  be given.

- (a) A *complete  $\varphi$ -type over  $A$*  is defined as in Definition 1.1.1, with  $b$  ranging over  $A$  instead of  $M$ ; such a type is *finitely satisfiable in  $A$*  if every finite subset of the type is realized by an element of  $A$ .
- (b) The type  $p$  is *definable over  $A$*  if  $p$  has a  $\varphi$ -definition that is an  $L(A)$ -formula.
- (c) The type  $p$  is *definable almost over  $A$*  if  $p$  has a  $\varphi$ -definition of the form  $\psi(y, c)$ , with  $\psi(y, z)$  an  $L$ -formula and  $c$  a tuple of  $N$ , such that only finitely many distinct subsets of  $N$  are of the form  $\psi(N, c')$  with  $c' \in N$  realizing the same complete type over  $A$  as  $c$ . In other words, the definition of  $p$  has only finitely many conjugates over  $A$ .

Definable over  $A$  implies definable almost over  $A$ : take  $c$  to enumerate the parameters from  $A$ , so that  $c' = c$  is the only possibility. The converse fails.

**Example 3.2.6.** Let  $T$  be the theory of an equivalence relation  $E$  with exactly two classes, both infinite, let  $\varphi(x, y)$  be  $E(x, y)$ , and let  $A = \emptyset$ . Let  $N$  be a model of  $T$  with classes  $C_1$  and  $C_2$ , and let  $p \in S_\varphi(N)$  be the  $\varphi$ -type saying that  $x$  lies in  $C_1$ , i.e.  $E(x, b) \in p$  if and only if  $b \in C_1$ . For any  $c \in C_1$ , the formula  $E(y, c)$  is a  $\varphi$ -definition of  $p$ . All elements of  $N$  have the same complete type over  $\emptyset$ , so the sets  $E(N, c')$  are exactly  $C_1$  and  $C_2$ . Hence  $p$  is definable almost over  $\emptyset$ . The type  $p$  is not definable over  $\emptyset$ : since  $N$  has an automorphism swapping  $C_1$  and  $C_2$ , every  $L$ -formula  $\psi(y)$  without parameters defines  $\emptyset$  or  $N$ , never  $C_1$ . Note that  $p$  is not finitely satisfiable in  $\emptyset$ .

### 3.3. Historical remarks

- ⌘ **Section 3.1.** The Ramsey-type argument that extracts ladders from trees is due to Hodges [53]. Hodges’s argument gives a direct combinatorial proof that tall trees force the order property, without going through definability of types. Conant and Terry [45] recently revisited the argument with explicit bounds, and extended the argument from  $\{0, 1\}$ -valued formulas to real-valued functions. Their version is the one followed here. Such quantitative bounds are what is needed for applications of stability to finite combinatorics, such as the regularity lemma for stable graphs.
- ⌘ **Section 3.2.** The rank  $R(\theta)$  is Shelah’s local rank  $R(\theta, \varphi, 2)$  [36, 57]. The proof that a type of minimal rank is definable is the classical route to definability of types, the one found in most textbook treatments, for instance Pillay [56] and Tent–Ziegler [58], and predates the combinatorial and analytic proofs given elsewhere in these notes. As noted above, the resulting definitions involve quantifiers, whereas those of Section 2.2 do not.



## CHAPTER 4

### Stability and notions of independence

Stability can also be understood through independence. In this chapter we show that the stable theories are exactly those carrying a notion of independence, written  $A \downarrow_M B$  and read “ $A$  is independent from  $B$  over  $M$ ”, that obeys a short list of natural axioms. The most important of these are finite character, symmetry, and the existence and uniqueness of independent extensions. The idea is that, given a type  $p$  over a model  $M$  and a set  $B$ , an independent extension of  $p$  to  $M \cup B$  adds no information beyond  $p$ . Stability is precisely the condition that such an extension always exists and is unique.

Both directions rest on Chapter 2. To obtain independence from stability we use definability of types: the independent extension is the *heir*, the extension of  $p$  given by the same definition scheme as  $p$ . The converse is a counting argument of the kind used there.

#### 4.1. Conventions

From now on  $T$  is a *complete* theory in the language  $L$ , and  $\kappa = |T| = |L| + \aleph_0$ . We work inside a monster model  $\mathbb{M}$  of  $T$ : a  $\bar{\kappa}$ -saturated and strongly  $\bar{\kappa}$ -homogeneous model for some very large cardinal  $\bar{\kappa}$  (larger than  $2^{2^\kappa}$ , say). A set is *small* if of cardinality  $< \bar{\kappa}$ . Letters  $A, B, C$  denote small subsets of  $\mathbb{M}$ ; tuples may be infinite, but are always small. We freely identify a set with an enumerating tuple. A *model* is a small elementary substructure  $M \preceq \mathbb{M}$ . Every model of  $T$  of size  $< \bar{\kappa}$  embeds elementarily in  $\mathbb{M}$ , and every type over a small set is realized in  $\mathbb{M}$ .

We write  $\text{tp}(a/A)$  for the complete type of  $a$  over  $A$ ,  $S_x(A)$  for the set of complete types over  $A$  in the variables  $x$ , and  $a \equiv_A a'$  for  $\text{tp}(a/A) = \text{tp}(a'/A)$ . We write  $MB$  for  $M \cup B$ . By strong homogeneity, we have  $a \equiv_A a'$  if and only if  $\sigma(a) = a'$  for some  $\sigma \in \text{Aut}(\mathbb{M}/A)$ .

**Definition 4.1.1.**

- (a) The theory  $T$  is *stable* if no  $L$ -formula  $\varphi(x, y)$  has the order property (see Definition 1.1.4).
- (b) For an infinite cardinal  $\lambda$ , the theory  $T$  is  $\lambda$ -*stable* if  $|S_x(A)| \leq \lambda$  whenever  $|A| \leq \lambda$  and  $x$  is a finite tuple of variables.

**Lemma 4.1.2.** *If  $T$  is  $\lambda$ -stable for some  $\lambda \geq \kappa$ , then  $T$  is stable.*

*Proof.* Suppose that  $\varphi(x, y)$  has the order property. By Proposition 2.3.1, the formula  $\varphi$  encodes binary trees of arbitrarily large finite height. By Proposition 2.4.2 there is a model  $N$  of  $T$  with  $|N| \leq \lambda$  and  $|S_\varphi(N)| > \lambda$ . Since  $T$  is complete, we may assume  $N \preceq \mathbb{M}$ . Every  $\varphi$ -type over  $N$  is consistent with the elementary diagram of  $N$ , so each  $\varphi$ -type over  $N$  extends to a complete type in  $S_x(N)$ , and distinct  $\varphi$ -types have distinct extensions. Hence  $|S_x(N)| > \lambda \geq |N|$ , and  $T$  is not  $\lambda$ -stable. ☺

## 4.2. Definitions and heirs

In this section we show that, in a stable theory, every type over a model has a definition scheme (see Lemma 4.2.1 below), use the scheme to extend the type to larger sets of parameters (the heir; see Definition 4.2.2 below), and prove that definitions over a model commute (see Lemma 4.2.4 below).

**Lemma 4.2.1.** *Assume that  $T$  is stable, and let  $p \in S_x(M)$  be given, where  $x$  may be infinite. For every  $L(M)$ -formula  $\varphi(x', y)$ , with  $x'$  a finite subtuple of  $x$  and  $y$  finite, there is an  $L(M)$ -formula  $d_p\varphi(y)$  such that for all  $b \in M$ , we have*

$$\varphi(x', b) \in p \iff M \models d_p\varphi(b).$$

Any two formulas with this property are equivalent in  $\mathbb{M}$ .

*Proof.* Write  $\varphi(x', y) = \varphi_0(x', y, m)$  with  $\varphi_0 \in L$  and  $m \in M$ . The set of formulas  $\varphi_0(x', c)^u$  ( $c \in M$ ) that belong to  $p$  is a complete  $\varphi_0$ -type over  $M$  in the sense of Definition 1.1.1. Since  $\varphi_0$  does not have the order property, Theorem 1.1.6 shows that this  $\varphi_0$ -type has a definition  $\psi(z)$  over  $M$ , which is a positive Boolean combination of instances of  $\varphi_0$ . Here  $z$  corresponds to the parameter variables  $y$  followed by those replaced by  $m$ . Then  $d_p\varphi(y) := \psi(y, m)$  works. If  $\psi_1(y)$  and  $\psi_2(y)$  both work, then  $M \models \forall y (\psi_1 \leftrightarrow \psi_2)$ . This is an  $L(M)$ -sentence, so this sentence also holds in  $\mathbb{M}$ .  $\text{☺}$

The family  $(d_p\varphi)_\varphi$  given by Lemma 4.2.1, determined up to equivalence in  $\mathbb{M}$ , is called the *definition scheme* of  $p$ .

**Definition 4.2.2.** Assume that  $T$  is stable. Let  $p \in S_x(M)$  be given, and let  $B$  be small. Define

$$p|B := \{\varphi(x', b) : \varphi(x', y) \in L(M), b \in B, \mathbb{M} \models d_p\varphi(b)\}.$$

The set  $p|B$  is called the *heir* of  $p$  over  $MB$ .

Every  $L(MB)$ -formula has the form  $\varphi(x', b)$  with  $\varphi \in L(M)$  and  $b \in B$ , so  $p|B$  is a set of  $L(MB)$ -formulas.

**Lemma 4.2.3.** *Assume that  $T$  is stable, let  $p \in S_x(M)$  be given, and let  $B$  and  $D$  be small sets. Then the following hold.*

- (a) *The set  $p|B$  belongs to  $S_x(MB)$  and extends  $p$ .*
- (b) *If  $B \subseteq D$ , then  $p|B$  is the restriction of  $p|D$  to  $MB$ .*
- (c) *If  $N$  is a model with  $M \subseteq N$ , then  $p|N$  has the same definition scheme as  $p$ , and  $(p|N)|B = p|NB$ .*

*Proof.* (a) *Completeness.* Let  $\varphi \in L(M)$  be given. Since  $p$  is complete, for every  $c \in M$  exactly one of  $\varphi(x', c)$ ,  $\neg\varphi(x', c)$  lies in  $p$ . Hence  $M \models \forall y (d_p(\neg\varphi)(y) \leftrightarrow \neg d_p\varphi(y))$ , and the same holds in  $\mathbb{M}$ . So for  $b \in B$  exactly one of  $\varphi(x', b)$ ,  $\neg\varphi(x', b)$  belongs to  $p|B$ .

*Consistency.* Let  $\varphi_i(x_i, b_i) \in p|B$  be given for  $i < n$ , and put  $\chi(\bar{x}; y_0, \dots, y_{n-1}) = \bigwedge_{i < n} \varphi_i(x_i, y_i)$ , where  $\bar{x}$  lists the variables in the  $x_i$ . For  $\bar{c} \in M$ , the formula  $\chi(\bar{x}, \bar{c})$  is in  $p$  if and only if each  $\varphi_i(x_i, c_i)$  is, and in that case  $\chi(\bar{x}, \bar{c})$  is satisfiable in  $M$ . Therefore

$$M \models \forall \bar{y} \left( \bigwedge_{i < n} d_p\varphi_i(y_i) \rightarrow d_p\chi(\bar{y}) \right) \quad \text{and} \quad M \models \forall \bar{y} (d_p\chi(\bar{y}) \rightarrow \exists \bar{x} \chi(\bar{x}, \bar{y})).$$

Both sentences transfer to  $\mathbb{M}$ . Applied to  $\bar{b} = (b_0, \dots, b_{n-1})$ , they give  $\mathbb{M} \models \exists \bar{x} \bigwedge_{i < n} \varphi_i(x_i, b_i)$ . By compactness, the set  $p|B$  is consistent.

*Extension.* For  $c \in M$  we have  $\varphi(x', c) \in p \iff M \models d_p\varphi(c)$ , which holds if and only if  $\varphi(x', c) \in p|B$ .

- (b) This is immediate from the definition.



(c) By (a), the type  $p|N$  extends  $p$ . For  $n \in N$  we have  $\varphi(x', n) \in p|N$  if and only if  $\mathbb{M} \models d_p\varphi(n)$ , so  $d_p\varphi$  is a definition of  $p|N$  for  $\varphi \in L(M)$ . For an  $L(N)$ -formula  $\varphi(x', y) = \varphi_0(x', y, n)$  with  $\varphi_0 \in L(M)$ , the formula  $d_p\varphi_0(y, n)$  defines  $p|N$ . By the uniqueness clause of Lemma 4.2.1, the definition scheme of  $p|N$  is thus given, up to equivalence, by that of  $p$ . Now for  $b \in B$  and  $n \in N$ , we have

$$\varphi_0(x', b, n) \in (p|N)|B \iff \mathbb{M} \models d_p\varphi_0(b, n) \iff \varphi_0(x', b, n) \in p|NB. \quad \odot$$

The heart of the matter is the following lemma, which says that definitions of types over a model commute; this lemma is the only place in this chapter where the absence of the order property is used directly. To state the lemma, let  $p \in S_x(M)$  and  $q \in S_y(M)$  be given, with  $x$  and  $y$  finite, and let  $\varphi(x, y) \in L(M)$  be given. As before, we write  $d_p\varphi(y)$  for the definition of  $p$  for  $\varphi(x, y)$ , and we write  $d_q\varphi(x)$  for the definition of  $q$  for  $\varphi$  regarded as a formula in the variables  $y$ , with parameter variables  $x$ .

**Lemma 4.2.4 (Harrington's lemma).** *Assume that  $T$  is stable, and let  $p \in S_x(M)$ ,  $q \in S_y(M)$  and  $\varphi(x, y) \in L(M)$  be given, where  $x$  and  $y$  are finite. Then*

$$d_p\varphi(y) \in q \iff d_q\varphi(x) \in p.$$

*Proof.* Choose recursively  $a_i, b_i \in \mathbb{M}$  ( $i < \omega$ ) such that  $a_i$  realizes  $p|\{b_j : j < i\}$  and  $b_i$  realizes  $q|\{a_j : j \leq i\}$ . This is possible by Lemma 4.2.3(a) and saturation. In particular  $a_i \models p$  and  $b_i \models q$ .

If  $j < i$ , then since  $a_i \models p|\{b_j\}$  and  $b_j \models q$ , we have

$$\mathbb{M} \models \varphi(a_i, b_j) \iff \mathbb{M} \models d_p\varphi(b_j) \iff d_p\varphi(y) \in q.$$

If  $i \leq j$ , then since  $b_j \models q|\{a_i\}$  and  $a_i \models p$ , we have

$$\mathbb{M} \models \varphi(a_i, b_j) \iff \mathbb{M} \models d_q\varphi(a_i) \iff d_q\varphi(x) \in p.$$

Suppose the two right-hand sides differ, and write  $\varphi(x, y) = \varphi_0(x, y, m)$  with  $\varphi_0 \in L$  and  $m \in M$ . Put  $b'_j = b_j m$ . Then either  $\mathbb{M} \models \varphi_0(a_i, b'_j) \iff i \leq j$  for all  $i, j$ , or  $\mathbb{M} \models \varphi_0(a_i, b'_j) \iff i > j$  for all  $i, j$ . In the first case  $\varphi_0(x; y, z)$  has the order property directly, and in the second by Remark 1.1.5. Either way this contradicts stability.  $\odot$



**Remark 4.2.5 (Coheirs, ultrafilters, and Harrington's lemma).** Harrington's lemma is the exchange of iterated ultrafilter limits of Chapter 1, in disguise. Let  $x$  and  $y$  be finite, and let  $\mathcal{U}$  be an ultrafilter on  $M^{|x|}$ . For a small set  $B$ , the *coheir* of  $\mathcal{U}$  over  $MB$  is the type

$$q_{\mathcal{U}}|B = \{\varphi(x, b) : \varphi(x, y) \in L(M), b \in B, \{a \in M^{|x|} : \mathbb{M} \models \varphi(a, b)\} \in \mathcal{U}\},$$

a complete type over  $MB$  that is finitely satisfiable in  $M$ ; every type over  $MB$  that is finitely satisfiable in  $M$  arises in this way. Let  $\mathcal{V}$  be an ultrafilter on  $M^{|y|}$ , let  $b$  realize  $q_{\mathcal{V}}|\emptyset$ , and let  $a$  realize  $q_{\mathcal{U}}|\{b\}$ . For  $\varphi(x, y) \in L(M)$ , unwinding the definitions gives

$$\mathbb{M} \models \varphi(a, b) \iff \lim_{a' \in \mathcal{U}} \lim_{b' \in \mathcal{V}} \varphi(a', b') = 1,$$

where  $\varphi(a', b')$  is read as 1 or 0. Realizing the two coheirs in the other order replaces this iterated limit by the one in the other order. So the two orders of realizing coheirs give the same type exactly when the iterated limits along  $\mathcal{U}$  and  $\mathcal{V}$  commute for every formula over  $M$ , and when  $T$  is stable they do, by Corollary 1.2.6 and Remark 1.2.2. In a stable theory, moreover, the coheir  $q_{\mathcal{U}}|B$  is the heir of  $q_{\mathcal{U}}|\emptyset$ , because both are nonforking extensions and nonforking extensions of types over models are unique (see Remark 4.4.2 and [56, 58]), so that  $\mathbb{M} \models d_p\varphi(b)$  holds exactly when  $\{a \in M^{|x|} : \mathbb{M} \models \varphi(a, b)\} \in \mathcal{U}$ . Read in this way,

Harrington's lemma, and with the lemma the symmetry of independence, is the commutation of ultrafilter limits.

### 4.3. Independence relations

In this section we list the axioms of an independence relation, show that when  $T$  is stable the heir relation satisfies all of them, and show that, conversely, a relation satisfying two of the axioms forces  $T$  to be stable (see Theorem 4.3.3 below).

**Definition 4.3.1.** An *independence relation (over models)* is a relation  $A \downarrow_M B$  between small sets  $A, B$  and models  $M$  satisfying the following axioms, for all small  $A, A', B, B'$  and all models  $M, N$ .

- (INV) *Invariance.* If  $A \downarrow_M B$  and  $\sigma \in \text{Aut}(\mathbb{M})$ , then  $\sigma(A) \downarrow_{\sigma(M)} \sigma(B)$ .
- (MON) *Monotonicity.* If  $A \downarrow_M B$ ,  $A' \subseteq A$  and  $B' \subseteq B$ , then  $A' \downarrow_M B'$ .
- (BMON) *Base monotonicity.* If  $A \downarrow_M B$  and  $N$  is a model with  $M \subseteq N \subseteq B$ , then  $A \downarrow_N B$ .
- (TR) *Transitivity.* If  $M \subseteq N$ ,  $A \downarrow_M N$  and  $A \downarrow_N B$ , then  $A \downarrow_M NB$ .
- (FC) *Finite character.* The relation  $A \downarrow_M B$  holds if and only if  $a \downarrow_M b$  for all finite tuples  $a$  from  $A$  and  $b$  from  $B$ .
- (SYM) *Symmetry.* The relation  $A \downarrow_M B$  holds if and only if  $B \downarrow_M A$ .
- (EX) *Existence (extension).* For all  $A, M, B$  there is  $A' \equiv_M A$  with  $A' \downarrow_M B$ .
- (UN) *Uniqueness (stationarity).* If  $A \equiv_M A'$ ,  $A \downarrow_M B$  and  $A' \downarrow_M B$ , then  $A \equiv_{MB} A'$ .
- (LOC) *Local character.* For every finite tuple  $a$  and every model  $N$  there is a model  $M \subseteq N$  with  $|M| \leq \kappa$  and  $a \downarrow_M N$ .

In terms of types, (EX) and (UN) say that a type  $p \in S_x(M)$  has exactly one *independent extension* to  $MB$ , namely the type over  $MB$  of any  $A \models p$  with  $A \downarrow_M B$ . All the other axioms are statements about which extensions of  $p$  count as independent.

**Definition 4.3.2.** Assume that  $T$  is stable. For small  $A, B$  and a model  $M$  we define

$$A \downarrow_M B \quad : \iff \quad \text{tp}(A/MB) = \text{tp}(A/M)|B.$$

In words, the type  $\text{tp}(A/MB)$  is the *heir* of  $\text{tp}(A/M)$ , the extension defined by the same definition scheme.

For finite tuples  $a, b$  with  $p = \text{tp}(a/M)$ , every  $L(Mb)$ -formula in  $x$  has the form  $\varphi(x, b)$  with  $\varphi(x, y) \in L(M)$ . So

$$(4.1) \quad a \downarrow_M b \iff \text{for all } \varphi(x, y) \in L(M) : \quad \mathbb{M} \models \varphi(a, b) \leftrightarrow d_p \varphi(b).$$

**Theorem 4.3.3.** For a complete theory  $T$  the following are equivalent.

- (1) The theory  $T$  is stable.
- (2) There is an independence relation satisfying all the axioms of Definition 4.3.1.
- (3) There is a relation  $A \downarrow_M B$  satisfying (UN) and (LOC).

Moreover, if  $T$  is stable, the relation of Definition 4.3.2 satisfies all the axioms. This relation also satisfies

- (AR) *Anti-reflexivity.* If  $a \downarrow_M a$  for a finite tuple  $a$ , then  $a \subseteq M$ .

**4.3.1. Proof of (1)  $\Rightarrow$  (2).** Assume that  $T$  is stable, and let  $\downarrow$  be the relation of Definition 4.3.2. Throughout, we write  $p = \text{tp}(A/M)$ .

(INV). Let  $\sigma \in \text{Aut}(\mathbb{M})$  be given. Applying  $\sigma$  to the parameters of the definition scheme of  $p$  gives a definition scheme for  $\sigma(p) = \text{tp}(\sigma A / \sigma M)$ . Hence  $\sigma(p)|\sigma(B) = \sigma(p|B)$ . If  $\text{tp}(A/MB) = p|B$ , then  $\text{tp}(\sigma A / \sigma M \sigma B) = \sigma(p|B) = \sigma(p)|\sigma B$ .

(MON). The definition scheme of  $\text{tp}(A'/M)$  is the part of that of  $p$  that involves only variables for  $A'$ . So  $\text{tp}(A'/M)|B'$  is the restriction of  $p|B$  to these variables and to  $MB'$  (see Lemma 4.2.3(b)), which is  $\text{tp}(A'/MB')$ .

(FC). The condition  $\text{tp}(A/MB) = p|B$  says that for every  $L(M)$ -formula  $\varphi(x', y)$  and every  $b \in B$  we have  $\mathbb{M} \models \varphi(a', b) \leftrightarrow d_p\varphi(b)$ , where  $a'$  is the finite subtuple of  $A$  corresponding to  $x'$ . Each instance involves only finite  $a' \subseteq A$  and  $b \subseteq B$ , and is exactly an instance of  $a' \perp_M b$ .

(BMON). Let  $N$  be a model with  $M \subseteq N \subseteq B$ , and suppose that  $\text{tp}(A/MB) = p|B$ . Then  $\text{tp}(A/N) = p|N$  by Lemma 4.2.3(b). By Lemma 4.2.3(c), we have  $\text{tp}(A/N)|B = (p|N)|B = p|NB = p|B = \text{tp}(A/NB)$ , so  $A \perp_N B$ .

(TR). Let  $N$  be a model with  $M \subseteq N$ , and suppose that  $\text{tp}(A/N) = p|N$  and  $\text{tp}(A/NB) = \text{tp}(A/N)|B$ . By Lemma 4.2.3(c), we have  $\text{tp}(A/NB) = (p|N)|B = p|NB$ , which means that  $A \perp_M NB$ .

(SYM). By (FC) we need only treat finite tuples  $a, b$ . Let  $p = \text{tp}(a/M)$  and  $q = \text{tp}(b/M)$ , and assume that  $a \perp_M b$ . For  $\varphi(x, y) \in L(M)$ , using (4.1) and Lemma 4.2.4, we have

$$\mathbb{M} \models \varphi(a, b) \iff \mathbb{M} \models d_p\varphi(b) \iff d_p\varphi(y) \in q \iff d_q\varphi(x) \in p \iff \mathbb{M} \models d_q\varphi(a).$$

By (4.1) with the roles of  $a$  and  $b$  exchanged, this says  $b \perp_M a$ .

(EX). By Lemma 4.2.3(a) and saturation, the type  $p|B$  has a realization  $A'$ . Then  $A' \equiv_M A$  and  $A' \perp_M B$ .

(UN). If  $A \equiv_M A'$ , then both have type  $p$  over  $M$ . If both are independent from  $B$  over  $M$ , then  $\text{tp}(A/MB) = p|B = \text{tp}(A'/MB)$ .

(LOC). Let  $a$  be finite, let  $N$  be a model, and let  $p = \text{tp}(a/N)$ . For each  $L$ -formula  $\varphi(x, y)$  fix a definition  $d_p\varphi(y) \in L(N)$ , and let  $D \subseteq N$  be the set of all parameters used, so  $|D| \leq \kappa$ . By downward Löwenheim–Skolem there is a model  $M \preceq N$  with  $D \subseteq M$  and  $|M| \leq \kappa$ . Let  $q = \text{tp}(a/M)$ . Each  $d_p\varphi$  is an  $L(M)$ -formula, and on  $M$  it agrees with the definition of  $p$ , so  $d_p\varphi$  defines  $q$  for  $\varphi$ . Hence these formulas form a definition scheme for  $q$  (for  $L(M)$ -formulas  $\varphi_0(x, y, m)$  use  $d_p\varphi_0(y, m)$ ). Hence  $q|N$  is defined by the same formulas as  $p$ . Since  $q|N$  and  $p$  are both complete types over  $N$ , we have  $q|N = p = \text{tp}(a/N)$ , which means that  $a \perp_M N$ .

(AR). Suppose that  $a \perp_M a$  with  $p = \text{tp}(a/M)$ , and suppose some coordinate  $a_0$  of  $a$  is not in  $M$ . Let  $\varphi(x, y)$  be the formula  $x_0 = y_0$ , where  $x_0$  and  $y_0$  are the corresponding coordinates of  $x$  and  $y$ . For  $c \in M$  we have  $\varphi(x, c) \notin p$ , because  $a_0 \neq c_0$ . So  $M \models \neg \exists y d_p\varphi(y)$ , and the same holds in  $\mathbb{M}$ . By (4.1), we have  $\mathbb{M} \models \varphi(a, a) \leftrightarrow d_p\varphi(a)$ . But  $\varphi(a, a)$  holds and  $d_p\varphi(a)$  fails, a contradiction.

**4.3.2. Proof of (2)  $\Rightarrow$  (3)  $\Rightarrow$  (1).** The implication (2)  $\Rightarrow$  (3) is immediate, because a relation satisfying all the axioms satisfies (UN) and (LOC) in particular. For (3)  $\Rightarrow$  (1), the idea is that (LOC) lets us describe each type over a model  $N$  by a type over a small submodel, and (UN) guarantees that this description loses no information; counting the descriptions then bounds the number of types. Let  $\perp$  satisfy (UN) and (LOC), and put  $\lambda = 2^\kappa$ , so that  $\lambda^\kappa = \lambda$ . We show that  $T$  is  $\lambda$ -stable. Then  $T$  is stable by Lemma 4.1.2.

Let  $A$  be a set with  $|A| \leq \lambda$ , and let  $x$  be a finite tuple of variables. By downward Löwenheim–Skolem there is a model  $N \supseteq A$  with  $|N| \leq \lambda$ , and restriction maps  $S_x(N)$  onto  $S_x(A)$ . So we need only show  $|S_x(N)| \leq \lambda$ .

For each  $p \in S_x(N)$  choose a realization  $a_p$ . By (LOC) choose a model  $M_p \subseteq N$  with  $|M_p| \leq \kappa$  and  $a_p \downarrow_{M_p} N$ . Consider the map

$$p \mapsto (M_p, \text{tp}(a_p/M_p)).$$

This map is injective. Indeed, suppose that  $M_p = M_{p'} =: M$  and  $a_p \equiv_M a_{p'}$ . Since  $a_p \downarrow_M N$  and  $a_{p'} \downarrow_M N$ , (UN) gives  $a_p \equiv_N a_{p'}$ , i.e.  $p = p'$ .

The number of subsets of  $N$  of size  $\leq \kappa$  is at most  $\lambda^\kappa = \lambda$ . For each such  $M$  we have  $|S_x(M)| \leq 2^\kappa = \lambda$ . Hence  $|S_x(N)| \leq \lambda \cdot \lambda = \lambda$ .  $\odot$

#### 4.4. Remarks

 **Remark 4.4.1.** The proof of (3)  $\Rightarrow$  (1) uses only uniqueness and local character. Local character may be weakened to  $|M| < \kappa_0$  for any fixed cardinal  $\kappa_0$ , by taking  $\lambda = 2^{\kappa_0 + \kappa}$ . So stability is equivalent to the existence of a relation for which types over models have unique independent extensions and every type is independent over a small submodel. All the other axioms come for free once  $T$  is stable.

 **Remark 4.4.2.** In a stable theory the relation of Definition 4.3.2 is *nonforking independence* restricted to models as bases. For  $M \subseteq B$ , the following are equivalent: the type  $\text{tp}(a/B)$  does not fork over  $M$ ; this type is the heir of  $\text{tp}(a/M)$ ; this type is definable over  $M$ ; see Pillay [56] or Tent–Ziegler [58]. Nonforking makes sense over arbitrary base sets  $C$ . Over such sets, nonforking satisfies all the axioms above except uniqueness, which holds only over algebraically closed sets in  $T^{\text{eq}}$  (Harnik–Harrington [54]). Kim and Pillay [55] proved that in a stable (indeed in a simple) theory, nonforking is the *only* relation on arbitrary small sets satisfying invariance, monotonicity, transitivity, finite character, symmetry, extension, local character and the independence theorem. Developing forking over arbitrary sets requires considerably more machinery, which is why we have taken models as base sets.

#### 4.5. Historical remarks

 **Abstract independence.** The axiomatic study of independence is older than model theory. Steinitz, in his foundational paper on the algebraic theory of fields [59], introduced transcendence bases and showed that any two transcendence bases have the same cardinality. The argument is an exchange argument of the kind now called the Steinitz exchange lemma, the same one that gives the dimension of a vector space. Van der Waerden, in *Moderne Algebra* [60], observed that linear dependence and algebraic dependence obey the same few rules, and derived the existence of bases and the invariance of their cardinality from an abstract notion of dependence whose central axiom is exchange. Whitney [61] went a step further. He introduced *matroids*, structures that isolate the abstract properties of linear dependence, and gave several equivalent axiomatizations, in terms of independent sets, bases, circuits and rank. Nakasawa developed the same theory independently in the mid-1930s, and Birkhoff [62] and Mac Lane [63] related matroid theory to lattices and to projective geometry.

After a period in which matroids were studied mainly by Rado and Tutte, Rota made them central to combinatorics. In his monograph with Crapo [64] they become *combinatorial geometries*, and a closure operator with the exchange property is called a *pregeometry*, the term still used in model theory. Rota’s view was that dependence is a geometric notion with invariants of its own, such as the Möbius function and the characteristic polynomial of the associated lattice.

Model theory met these ideas in the study of uncountably categorical theories. Baldwin and Lachlan [65] showed that in a strongly minimal set algebraic closure has the exchange

property, so algebraic closure is a pregeometry and every model has a well-defined dimension. They used this to give a new proof of Morley’s categoricity theorem [35]. Shelah’s forking [57] carries independence far beyond pregeometries: nonforking independence is not given by a closure operator, yet in a stable theory nonforking independence has the symmetry, existence and uniqueness properties on which arguments about dimension depend. The later work of Zilber and Hrushovski on the geometry of forking turned the study of forking into a subject of its own, geometric stability theory [56]. The point of view of this chapter, which characterizes independence axiomatically, as van der Waerden and Whitney did for linear dependence, is that of Harnik and Harrington and of Kim and Pillay, discussed below.

- ✂ **Section 4.1.** Working inside a single large saturated and strongly homogeneous model goes back to Shelah [57], who wrote  $\mathfrak{C}$  for this model. The name *monster model* became standard later. The notion of  $\omega$ -stability comes from Morley’s categoricity theorem [35]. The study of  $\lambda$ -stability for arbitrary  $\lambda$ , including the fact that stability is equivalent to  $\lambda$ -stability for all  $\lambda$  with  $\lambda^{|T|} = \lambda$ , is due to Shelah [36, 57].
- ✂ **Section 4.2.** Heirs and coheirs (*héritiers* and *cohéritiers*) were introduced by Lascar and Poizat [33]. They gave an alternative presentation of Shelah’s forking in stable theories, based only on definability of types and finite satisfiability. Lemma 4.2.4, which says that the definitions of two types over a model commute, is due to Harrington. The lemma appears in Harnik–Harrington [54], and is the key step in proving symmetry of independence in stable theories.
- ✂ **Section 4.3.** Nonforking independence was introduced by Shelah [57]. Lascar and Poizat [33] reformulated nonforking independence in terms of heirs and definability. The characterization of stability through abstract independence notions is due to Harnik and Harrington [54]. Kim and Pillay [55] later extended this characterization from stable to simple theories. They proved that the simple theories are exactly those carrying a relation satisfying the axioms listed above, with the independence theorem in place of uniqueness, and that such a relation is then nonforking. Theorem 4.3.3 is the stable case over models, with uniqueness (stationarity over models) as the key axiom. The anchor symbol  $\perp$  for independence was introduced by Makkai in a lecture in Jerusalem, during the Model Theory Year at the Institute for Advanced Studies (1980–81), and was adopted at once. The symbol appears in print in Makkai’s survey [48].



## CHAPTER 5

### From stability to NIP

In a stable theory no formula can encode arbitrarily long ladders. Many of the best understood structures in mathematics fail this test for a simple reason: they carry a definable linear order. Dense linear orders, the ordered field of real numbers and, more generally, o-minimal structures are all unstable, and yet their definable sets are remarkably well behaved. To account for this kind of tameness we introduce the *independence property*, a configuration much wilder than a ladder, and study the formulas and theories that avoid the independence property, called *NIP*. Every stable formula is NIP, and so are the orders just mentioned, so NIP generalizes both stability and the tameness of order. In Section 5.1 we keep the conventions of Section 1.1, and in Section 5.2 those of Section 1.2.

#### 5.1. The independence property

A ladder (see Definition 1.1.4) is a special pattern: the element  $a_i$  satisfies  $\varphi(x, b_j)$  exactly for  $j \geq i$ . The independence property asks instead that *every* pattern be realized.

**Definition 5.1.1.** Let  $M$  be a model of  $T$ , and let  $n \geq 1$  be given.

- (a) Tuples  $b_1, \dots, b_n$  of  $M$  are *shattered by  $\varphi$  in  $M$*  if for every  $S \subseteq \{1, \dots, n\}$  there is a tuple  $a_S$  of  $M$  such that, for all  $1 \leq i \leq n$ , we have

$$M \models \varphi(a_S, b_i) \iff i \in S.$$

- (b) The formula  $\varphi$  has the *independence property* if for every  $n$  some model of  $T$  contains  $n$  tuples shattered by  $\varphi$ . Otherwise  $\varphi$  is *NIP* (“not the independence property”); NIP formulas are also called *dependent*.
- (c) The *VC dimension* of  $\varphi$  is the largest  $n$  such that some model of  $T$  contains  $n$  tuples shattered by  $\varphi$ , and  $\infty$  if there is no largest such  $n$ .

A theory is *NIP* if every formula is NIP.

Thus  $\varphi$  is NIP if and only if the VC dimension of  $\varphi$  is finite. The terminology comes from learning theory. For  $a \in M$  we write  $\varphi(a, M) = \{b \in M : M \models \varphi(a, b)\}$ . If  $b_1, \dots, b_n$  are shattered by  $\varphi$  in  $M$ , then every subset of  $\{b_1, \dots, b_n\}$  is cut out by one of the sets  $\varphi(a, M)$ ; in the language of learning theory, the family  $\{\varphi(a, M) : a \in M\}$  shatters  $\{b_1, \dots, b_n\}$ . As for ladders, an  $(n+1)$ -shattered family contains an  $n$ -shattered one, so by compactness  $\varphi$  has the independence property if and only if some model of  $T$  contains an infinite sequence  $(b_i)_{i < \omega}$  such that for every  $S \subseteq \omega$  there is  $a_S$  with  $\varphi(a_S, b_i)$  if and only if  $i \in S$ .

**Proposition 5.1.2.** *If  $\varphi$  does not have the order property, then  $\varphi$  is NIP.*

*Proof.* Suppose that  $b_1, \dots, b_n$  are shattered by  $\varphi$  in  $M$ , witnessed by the tuples  $a_S$ . For  $1 \leq i \leq n$  put  $S_i = \{j : i \leq j \leq n\}$  and  $a_i = a_{S_i}$ . Then  $M \models \varphi(a_i, b_j)$  if and only if  $j \in S_i$ , that is, if and only if  $i \leq j$ . So  $a_1, \dots, a_n$  and  $b_1, \dots, b_n$  form an  $n$ -ladder. If  $\varphi$  has the independence property, then  $\varphi$  also has the order property. ☺

The converse fails, and the first example of the chapter shows why NIP is the right notion for ordered structures.



**Example 5.1.3.** Let  $T$  be the theory of  $(\mathbb{Q}, <)$  and let  $\varphi(x, y)$  be  $x < y$ . Then  $a_i = i$  and  $b_j = j + \frac{1}{2}$  satisfy  $\varphi(a_i, b_j)$  if and only if  $i \leq j$ , so  $\varphi$  has the order property and is unstable. But  $\varphi$  is NIP, with VC dimension 1. Indeed, let  $b_1, b_2$  be elements of a model, say  $b_1 \leq b_2$ . Shattering requires some  $a$  with  $a < b_1$  and  $a \not< b_2$ , that is, some  $a$  with  $b_2 \leq a < b_1$ , which is impossible. So no two elements are shattered. A single element  $b$  is shattered, by any  $a < b$  and any  $a \geq b$ .

**Example 5.1.4.** Let  $T$  be the theory of the random graph, whose models are the graphs in which, for all disjoint finite sets  $U, V$  of vertices, some vertex outside  $U \cup V$  is adjacent to every vertex of  $U$  and to no vertex of  $V$ . Let  $\varphi(x, y)$  be the edge relation  $E(x, y)$ . Any  $n$  distinct vertices  $b_1, \dots, b_n$  are shattered: for  $S \subseteq \{1, \dots, n\}$  apply the extension property to  $U = \{b_i : i \in S\}$  and  $V = \{b_i : i \notin S\}$ . So  $E$  has the independence property.

The independence property of  $\varphi(x, y)$  refers to shattering tuples on the  $y$ -side. In Proposition 5.1.5 below we show that the choice of side does not matter. We write  $\varphi^{\text{opp}}(y, x)$  for  $\varphi(x, y)$  with the roles of the variables exchanged.

**Proposition 5.1.5.** *The formula  $\varphi$  is NIP if and only if  $\varphi^{\text{opp}}$  is NIP. More precisely, if some model contains  $2^n$  tuples shattered by  $\varphi$ , then that model contains  $n$  tuples shattered by  $\varphi^{\text{opp}}$ .*

*Proof.* Index the  $2^n$  tuples as  $b_T$  with  $T \subseteq \{1, \dots, n\}$ , and let the tuples  $a_S$  with  $S \subseteq \mathcal{P}(\{1, \dots, n\})$  witness that they are shattered by  $\varphi$ . For  $1 \leq i \leq n$  put  $S_i = \{T : i \in T\}$  and  $a_i = a_{S_i}$ . Then  $M \models \varphi(a_i, b_T)$  if and only if  $T \in S_i$ , that is, if and only if  $i \in T$ . So for every  $T \subseteq \{1, \dots, n\}$  the tuple  $b_T$  satisfies  $\varphi^{\text{opp}}(b_T, a_i)$  exactly for  $i \in T$ , and  $a_1, \dots, a_n$  are shattered by  $\varphi^{\text{opp}}$ . Applying this in both directions gives the equivalence.  $\text{☺}$



**Remark 5.1.6.** A formula  $\psi(x, y)$  has the *strict order property* if for every  $n$  some model  $M$  of  $T$  contains tuples  $b_1, \dots, b_n$  with  $\psi(M, b_1) \subsetneq \psi(M, b_2) \subsetneq \dots \subsetneq \psi(M, b_n)$ , where  $\psi(M, b) = \{a \in M : M \models \psi(a, b)\}$ . The formula  $x < y$  of Example 5.1.3 is the basic case. Shelah proved that a theory is unstable if and only if some formula has the independence property or some formula has the strict order property [57, Chapter II]. So an unstable theory either encodes arbitrary subsets, as the random graph does, or encodes an infinite order; NIP theories are those in which only the second reason can occur.

## 5.2. The real-valued case

As in Section 1.2, we now replace formulas by functions on arbitrary sets. On arbitrary sets compactness is not available, so shattering is required only for finite patterns.

**Definition 5.2.1.** Let  $A, B$  be sets and  $\varphi : A \times B \rightarrow \mathbb{R}$  a bounded function. A sequence  $(b_n)_{n < \omega}$  in  $B$  is *independent for  $\varphi$*  if there are reals  $r < s$  such that for all disjoint finite sets  $P, Q \subseteq \omega$  there is  $a \in A$  with

$$\varphi(a, b_n) \geq s \text{ for } n \in P \quad \text{and} \quad \varphi(a, b_n) \leq r \text{ for } n \in Q.$$

The function  $\varphi$  has the *independence property* if some sequence in  $B$  is independent for  $\varphi$ , and  $\varphi$  is *NIP* otherwise.

For a  $\{0, 1\}$ -valued function, any  $0 < r < s < 1$  may be used, and a sequence is independent exactly when each finite part of the sequence is shattered in the sense of Definition 5.1.1. By compactness, a formula  $\varphi(x, y)$  is NIP in the sense of Definition 5.1.1 if and only if the corresponding function on  $M^{|x|} \times M^{|y|}$  is NIP for every model  $M$  of  $T$ .

Stability was characterized in Section 1.2 by the exchange of iterated limits. NIP is characterized by the existence of convergent subsequences. For  $b \in B$  let  $e_b : \text{LS}(\varphi) \rightarrow [0, 1]$  be the evaluation map  $e_b(p) = p(b)$ , as in Subsection 1.2.4. The analytic input is Rosenthal's dichotomy.



**Fact 5.2.2 (Rosenthal).** *Let  $X$  be a set and  $(f_n)$  a uniformly bounded sequence of real-valued functions on  $X$ . Then  $(f_n)$  has a subsequence that converges pointwise on  $X$ , or a subsequence  $(f_{n_k})$  for which there are reals  $r < s$  such that for all disjoint finite sets  $P, Q \subseteq \omega$  there is  $x \in X$  with  $f_{n_k}(x) \geq s$  for  $k \in P$  and  $f_{n_k}(x) \leq r$  for  $k \in Q$ .*

**Theorem 5.2.3.** *For  $\varphi: A \times B \rightarrow [0, 1]$  the following are equivalent.*

- (1) *The function  $\varphi$  is NIP.*
- (2) *For every sequence  $(b_n)$  in  $B$ , the sequence of functions  $(e_{b_n})$  has a subsequence that converges pointwise on  $\text{LS}(\varphi)$ .*

*Proof.* (2)  $\Rightarrow$  (1). Suppose that  $(b_n)$  is independent for  $\varphi$ , with reals  $r < s$ , and let  $(b_{n_k})$  be a subsequence. For each  $K < \omega$  choose  $a_K \in A$  with  $\varphi(a_K, b_{n_k}) \geq s$  for even  $k < K$  and  $\varphi(a_K, b_{n_k}) \leq r$  for odd  $k < K$ . Let  $\mathcal{U}$  be a nonprincipal ultrafilter on  $\omega$  and  $p = \lim_{K, \mathcal{U}} \text{ltp}_\varphi(a_K)$ , a point of  $\text{LS}(\varphi)$ . For each fixed  $k$  the inequality for  $k$  holds for all  $K > k$ , so  $e_{b_{n_k}}(p) = p(b_{n_k})$  is at least  $s$  for even  $k$  and at most  $r$  for odd  $k$ . Hence  $(e_{b_{n_k}}(p))_k$  does not converge. Since the subsequence was arbitrary, (2) fails.

(1)  $\Rightarrow$  (2). Suppose (2) fails for  $(b_n)$ . By Fact 5.2.2 applied to  $X = \text{LS}(\varphi)$  and  $f_n = e_{b_n}$ , there are a subsequence  $(b_{n_k})$  and reals  $r < s$  such that for all disjoint finite  $P, Q \subseteq \omega$  there is  $p \in \text{LS}(\varphi)$  with  $p(b_{n_k}) \geq s$  for  $k \in P$  and  $p(b_{n_k}) \leq r$  for  $k \in Q$ . Fix  $r' < s'$  with  $r < r' < s' < s$ . Since  $p$  is a pointwise limit of left types  $\text{ltp}_\varphi(a)$  and only finitely many coordinates are involved, some  $a \in A$  satisfies  $\varphi(a, b_{n_k}) > s'$  for  $k \in P$  and  $\varphi(a, b_{n_k}) < r'$  for  $k \in Q$ . So  $(b_{n_k})$  is independent for  $\varphi$ , with  $r' < s'$ , and  $\varphi$  is not NIP.  $\text{☹}$

The failure of condition (2) has a striking topological form: an independent sequence produces a copy of  $\beta\mathbb{N}$ .

**Proposition 5.2.4.** *If  $(b_n)$  is independent for  $\varphi$ , then the map  $\mathcal{U} \mapsto \lim_{n, \mathcal{U}} e_{b_n}$  is a homeomorphism from  $\beta\mathbb{N}$  onto a closed subset of the pointwise closure of  $\{e_b : b \in B\}$  in  $[0, 1]^{\text{LS}(\varphi)}$ .*

*Proof.* The map  $n \mapsto e_{b_n}$  from  $\mathbb{N}$  into the compact space  $[0, 1]^{\text{LS}(\varphi)}$  extends to the continuous map  $\mathcal{U} \mapsto \lim_{n, \mathcal{U}} e_{b_n}$  on  $\beta\mathbb{N}$ , whose image lies in the pointwise closure of  $\{e_b : b \in B\}$ . Since  $\beta\mathbb{N}$  is compact and  $[0, 1]^{\text{LS}(\varphi)}$  is Hausdorff, it remains to be shown that the map is injective. Let  $r < s$  witness the independence of  $(b_n)$ , let  $\mathcal{U} \neq \mathcal{V}$  be ultrafilters on  $\mathbb{N}$ , and choose  $X \in \mathcal{U}$  with  $\mathbb{N} \setminus X \in \mathcal{V}$ . For each finite  $F \subseteq \mathbb{N}$  there is  $a_F \in A$  with  $\varphi(a_F, b_n) \geq s$  for  $n \in F \cap X$  and  $\varphi(a_F, b_n) \leq r$  for  $n \in F \setminus X$ . Let  $\mathcal{D}$  be an ultrafilter on the finite subsets of  $\mathbb{N}$  that contains  $\{G : G \supseteq F\}$  for every finite  $F$ , and put  $p = \lim_{F, \mathcal{D}} \text{ltp}_\varphi(a_F) \in \text{LS}(\varphi)$ . Then  $p(b_n) \geq s$  for  $n \in X$  and  $p(b_n) \leq r$  for  $n \notin X$ . Hence the limit of  $e_{b_n}(p)$  along  $\mathcal{U}$  is at least  $s$ , the limit along  $\mathcal{V}$  is at most  $r$ , and the two limits of  $(e_{b_n})$  differ at  $p$ .  $\text{☹}$

So the instances of  $\varphi$ , viewed as continuous functions on  $\text{LS}(\varphi)$ , face a dichotomy. If  $\varphi$  is NIP, every sequence of instances has a pointwise convergent subsequence (see Theorem 5.2.3); if  $\varphi$  has the independence property, the closure of the instances contains  $\beta\mathbb{N}$ , a compact space in which no sequence of distinct points converges. This is the topological face of Rosenthal's  $\ell_1$  dichotomy. In the theory of Bourgain, Fremlin and Talagrand [1], the same dichotomy separates the *Rosenthal compacta*, the compact spaces of Baire class 1 functions on Polish spaces, from compact spaces of functions that contain  $\beta\mathbb{N}$ , and Simon [14] develops the NIP side of this picture for spaces of types. Stable functions sit at the rigid end of the spectrum: their type spaces are Eberlein compacta (see Corollary A.1.7 below).

The proof of (2)  $\Rightarrow$  (1) above is elementary; only (1)  $\Rightarrow$  (2) needs Rosenthal's dichotomy. The next proposition shows directly, using only the separately continuous extension of Proposition 1.2.8, that stability implies NIP also in the real-valued setting.

**Proposition 5.2.5.** *Every stable function  $\varphi: A \times B \rightarrow [0, 1]$  is NIP.*

*Proof.* Let  $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$  be the separately continuous function of Proposition 1.2.8. Taking a constant family in (1.2) gives  $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = p(b)$  for all  $p \in \text{LS}(\varphi)$  and  $b \in B$ .

Suppose, towards a contradiction, that  $(b_n)$  is independent for  $\varphi$ , with reals  $r < s$ . For  $m < N$  choose  $a_{m,N} \in A$  with

$$\varphi(a_{m,N}, b_n) \geq s \text{ for } m \leq n < N \quad \text{and} \quad \varphi(a_{m,N}, b_n) \leq r \text{ for } n < m,$$

and let  $a_{m,N}$  be arbitrary for  $N \leq m$ . Fix a nonprincipal ultrafilter  $\mathcal{U}$  on  $\omega$  and put

$$p_m = \lim_{N, \mathcal{U}} \text{ltp}_\varphi(a_{m,N}) \in \text{LS}(\varphi).$$

For fixed  $n$  the displayed inequalities hold for all  $N > \max(m, n)$ , so

$$(5.1) \quad p_m(b_n) \geq s \text{ if } n \geq m, \quad p_m(b_n) \leq r \text{ if } n < m.$$

Now put  $q = \lim_{n, \mathcal{U}} \text{rtp}_\varphi(b_n) \in \text{RS}(\varphi)$  and  $p = \lim_{m, \mathcal{U}} p_m \in \text{LS}(\varphi)$ . We compute  $\hat{\varphi}(p, q)$  in two ways.

First, by continuity of  $\hat{\varphi}(p_m, \cdot)$  and (5.1), we have  $\hat{\varphi}(p_m, q) = \lim_{n, \mathcal{U}} p_m(b_n) \geq s$  for every  $m$ , since  $p_m(b_n) \geq s$  for all  $n \geq m$ . By continuity of  $\hat{\varphi}(\cdot, q)$ , we obtain  $\hat{\varphi}(p, q) = \lim_{m, \mathcal{U}} \hat{\varphi}(p_m, q) \geq s$ .

Second, since limits in  $\text{LS}(\varphi)$  are computed coordinatewise, we have  $p(b_n) = \lim_{m, \mathcal{U}} p_m(b_n) \leq r$  for every  $n$ , because  $p_m(b_n) \leq r$  for all  $m > n$ . By continuity of  $\hat{\varphi}(p, \cdot)$ , we obtain  $\hat{\varphi}(p, q) = \lim_{n, \mathcal{U}} \hat{\varphi}(p, \text{rtp}_\varphi(b_n)) = \lim_{n, \mathcal{U}} p(b_n) \leq r$ .

So  $s \leq \hat{\varphi}(p, q) \leq r$ , contradicting  $r < s$ . ☺

The types  $p_m$  in the proof of Proposition 5.2.5 form a ladder in  $\text{LS}(\varphi)$  against the  $b_n$ , which is exactly what the separate continuity of  $\hat{\varphi}$  forbids: an independent sequence is shattered in every pattern, and a staircase is one of the patterns.

 **Remark 5.2.6.** Proposition 5.2.5 can also be derived, with heavier machinery, from Theorem 5.2.3. By Grothendieck's double limit criterion, in the form of Proposition A.1.6, stability of  $\varphi$  makes  $\{e_b : b \in B\}$  relatively weakly compact in  $C(\text{LS}(\varphi))$ . By Corollary A.1.5, which rests on the Eberlein–Šmulian theorem (see Fact A.1.4 below), every sequence  $(e_{b_n})$  then has a pointwise convergent subsequence. Thus condition (2) of Theorem 5.2.3 holds, and  $\varphi$  is NIP by the implication (2)  $\Rightarrow$  (1), which does not use Rosenthal's dichotomy. The direct proof above avoids both Grothendieck's criterion and the Eberlein–Šmulian theorem.

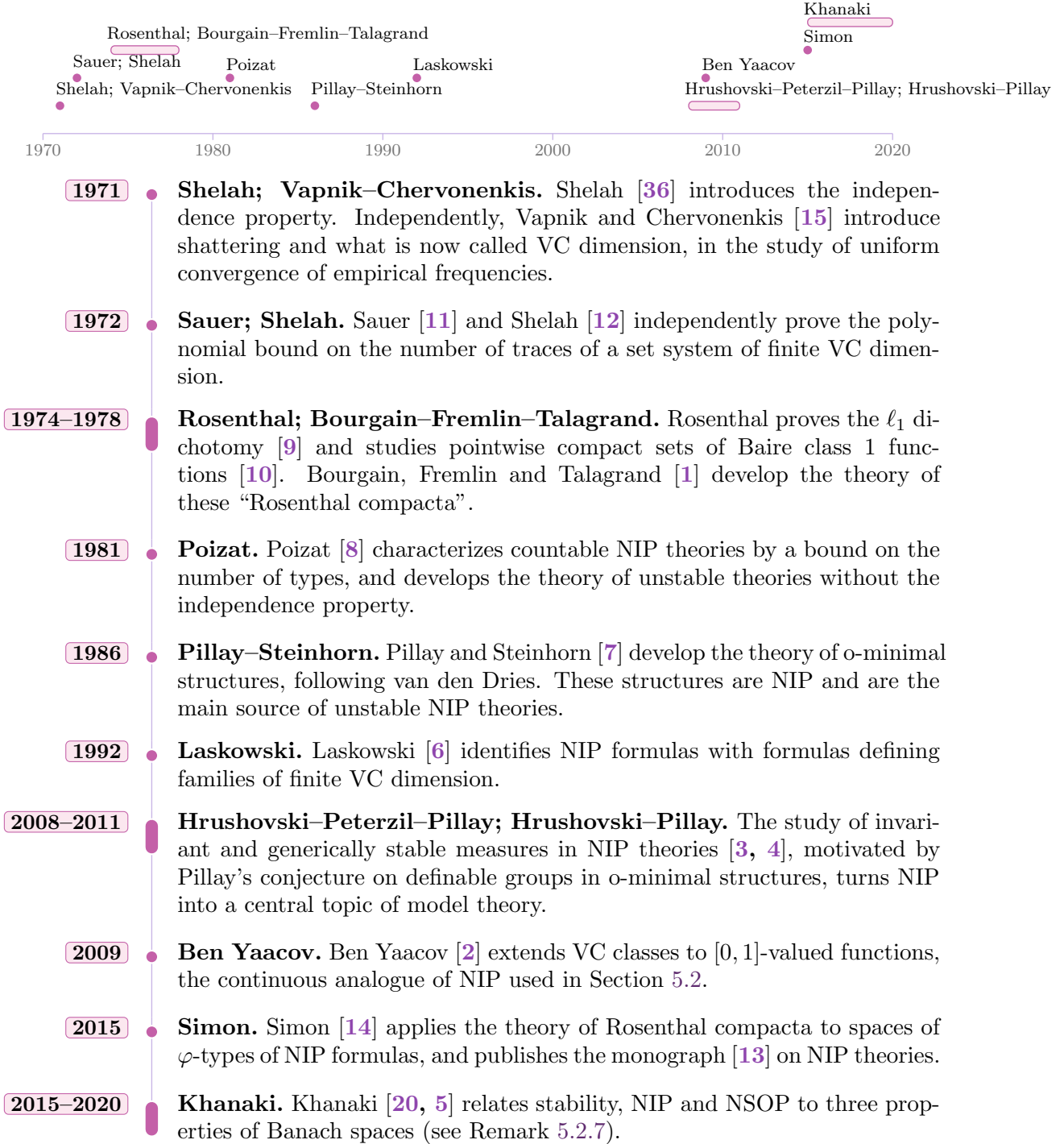
 **Remark 5.2.7.** Theorem 1.2.9 represents stable functions by the pairing of a reflexive Banach space  $E$  with  $E^*$ . Khanaki [5] proved that  $\varphi$  is stable, NIP, or NSOP (that is, without the strict order property of Remark 5.1.6) according as a Banach space naturally associated with  $\varphi$  is reflexive, contains no copy of  $\ell_1$ , or is weakly sequentially complete (every weakly Cauchy sequence converges weakly).

### 5.3. Historical remarks

 **Section 5.1.** The independence property was introduced by Shelah [36], in the same paper as the order property. His theorem that an unstable theory has the independence property or the strict order property is in [57, Chapter II]. The combinatorial notion of shattering, and the VC dimension, were introduced independently in probability and learning theory by Vapnik and Chervonenkis [15]; the polynomial bound on the number of traces (the subsets cut out on an  $n$ -element set) of a family of finite VC dimension was found independently by Sauer [11] and by Shelah [12]. Laskowski [6] observed that a formula is NIP exactly when

the family of sets defined by the formula has finite VC dimension, connecting model theory with learning theory.

✂ **Section 5.2.** Rosenthal’s dichotomy comes from his characterization of Banach spaces containing  $\ell_1$  [9], and the systematic study of pointwise compact sets of Baire class 1 functions is due to Rosenthal [10] and to Bourgain, Fremlin and Talagrand [1]. Their work entered model theory much later. The development is summarized in the following timeline.





## APPENDIX A

### Grothendieck's double limit criterion

Grothendieck's double limit criterion is used at several points in these notes: in Subsection 1.2.4 and in Remarks 1.2.14 and 5.2.6. This appendix states the criterion in the form we need and explains how the criterion relates to the stability of functions (see Definition 1.2.5). Throughout, we fix a compact Hausdorff space  $K$ , and  $C(K)$  denotes the Banach space of continuous real-valued functions on  $K$  with the supremum norm. The *weak topology* on  $C(K)$  is the coarsest topology making every bounded linear functional continuous, and the *pointwise topology* is the one induced by the product topology of  $\mathbb{R}^K$ . Point evaluations  $f \mapsto f(x)$  are bounded linear functionals, so the weak topology is finer than the pointwise topology.

#### A.1. The criterion

**Definition A.1.1.** A bounded set  $H \subseteq C(K)$  satisfies the *double limit condition* if for all sequences  $(x_m)$  in  $K$  and  $(f_n)$  in  $H$  and all ultrafilters  $\mathcal{U}, \mathcal{V}$  on  $\mathbb{N}$ , we have

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} f_n(x_m) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} f_n(x_m).$$

In the language of Definition 1.2.5, the double limit condition says that the evaluation function  $K \times H \rightarrow \mathbb{R}$ ,  $(x, f) \mapsto f(x)$ , is stable. Grothendieck stated the condition with ordinary limits; the next lemma shows that the two formulations are equivalent.

**Lemma A.1.2.** *A bounded set  $H \subseteq C(K)$  satisfies the double limit condition if and only if*

$$\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$$

*for all sequences  $(x_m)$  in  $K$  and  $(f_n)$  in  $H$  for which both iterated limits exist.*

*Proof.* If both ordinary iterated limits exist, these limits are equal to the iterated limits along any nonprincipal ultrafilters, so the double limit condition implies the second condition.

Conversely, suppose the double limit condition fails for  $(x_m)$ ,  $(f_n)$ ,  $\mathcal{U}$  and  $\mathcal{V}$ , with iterated limits  $\alpha \neq \beta$ . After an affine change of scale we may assume that all values lie in  $[0, 1]$ . Proposition 1.2.3, applied to the function  $(x, f) \mapsto f(x)$ , gives subsequences, which we still denote  $(x_m)$  and  $(f_n)$ , with  $|f_n(x_m) - \alpha| \leq 2^{1-m}$  for  $m < n$  and  $|f_n(x_m) - \beta| \leq 2^{1-n}$  for  $n < m$ . By a diagonal argument we may pass to further subsequences along which  $\lim_n f_n(x_m)$  exists for every  $m$  and  $\lim_m f_n(x_m)$  exists for every  $n$ ; the inequalities are preserved, with the indices renumbered. Then  $\lim_n f_n(x_m)$  is within  $2^{1-m}$  of  $\alpha$ , so  $\lim_m \lim_n f_n(x_m) = \alpha$ , and symmetrically  $\lim_n \lim_m f_n(x_m) = \beta$ . So the second condition fails. ☺

**Theorem A.1.3 (Grothendieck).** *Let  $H \subseteq C(K)$  be bounded. The following are equivalent.*

- (1) *The set  $H$  is relatively weakly compact in  $C(K)$ .*
- (2) *Every function in the pointwise closure of  $H$  in  $\mathbb{R}^K$  is continuous.*
- (3) *The set  $H$  satisfies the double limit condition.*

*Proof.* (2)  $\Rightarrow$  (3). Let  $(x_m)$ ,  $(f_n)$ ,  $\mathcal{U}$ ,  $\mathcal{V}$  be given, and put  $x = \lim_{m,\mathcal{U}} x_m \in K$  and  $f = \lim_{n,\mathcal{V}} f_n \in \mathbb{R}^K$ . By (2), the function  $f$  is continuous. Using the continuity of  $f$  and of each  $f_n$ , we obtain

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} f_n(x_m) = \lim_{m,\mathcal{U}} f(x_m) = f(x) = \lim_{n,\mathcal{V}} f_n(x) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} f_n(x_m).$$

(3)  $\Rightarrow$  (2). Suppose that  $f$  is in the pointwise closure of  $H$  but is not continuous at some  $x \in K$ . Then  $f = \lim_{i,\mathcal{U}} f_i$  for a family  $(f_i)_{i \in I}$  in  $H$  and an ultrafilter  $\mathcal{U}$  on  $I$ , and there are a family  $(x_j)_{j \in J}$  in  $K$  and an ultrafilter  $\mathcal{V}$  on  $J$  with  $\lim_{j,\mathcal{V}} x_j = x$  and  $\lim_{j,\mathcal{V}} f(x_j) = \beta \neq f(x)$ . (Fix  $\varepsilon > 0$  such that every neighborhood  $O$  of  $x$  contains a point  $x_O$  with  $|f(x_O) - f(x)| \geq \varepsilon$ , and let  $\mathcal{V}$  be an ultrafilter on the set of neighborhoods of  $x$  containing, for each  $O$ , the set of neighborhoods contained in  $O$ .) By continuity of each  $f_i$ , we have

$$\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} f_i(x_j) = \lim_{i,\mathcal{U}} f_i(x) = f(x), \quad \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} f_i(x_j) = \lim_{j,\mathcal{V}} f(x_j) = \beta.$$

After an affine change of scale, Proposition 1.2.3 and the proof of Corollary 1.2.6 turn these families into sequences whose iterated limits along a nonprincipal ultrafilter are  $\beta$  and  $f(x)$ . This contradicts (3).

(1)  $\Rightarrow$  (2). Let  $W$  be the weak closure of  $H$ , which is weakly compact by (1). The identity map from  $C(K)$  with the weak topology to  $C(K)$  with the pointwise topology is continuous, so  $W$  is pointwise compact, hence closed in  $\mathbb{R}^K$ . So the pointwise closure of  $H$  is contained in  $W \subseteq C(K)$ .

(2)  $\Rightarrow$  (1). Let  $\bar{H}$  be the pointwise closure of  $H$ . This set is compact in  $\mathbb{R}^K$ , because  $H$  is bounded, and is contained in  $C(K)$  by (2). We need only show that the weak and pointwise topologies agree on  $\bar{H}$ , since then  $\bar{H}$  is weakly compact. By the Riesz representation theorem, the weak topology is generated by the maps  $h \mapsto \int h d\mu$  with  $\mu$  a Radon measure on  $K$ . On the uniformly bounded set  $\bar{H}$  each such map is sequentially continuous for the pointwise topology, by Lebesgue's dominated convergence theorem. Grothendieck proved that a pointwise compact subset of  $C(K)$  is *angelic*: every point in the closure of a subset is the limit of a sequence from that subset [44]; see also Pryce [25]. On an angelic space sequentially continuous functions are continuous, which completes the proof.  $\odot$

The proof of (2)  $\Leftrightarrow$  (3) in Theorem A.1.3 uses only ultrafilter limits and Proposition 1.2.3. The measure-theoretic input and the angelicity of pointwise compact sets enter only in (2)  $\Rightarrow$  (1).

**Fact A.1.4 (Eberlein–Šmulian).** *In a Banach space, every sequence in a relatively weakly compact set has a weakly convergent subsequence [16, 17].*

**Corollary A.1.5.** *If a bounded set  $H \subseteq C(K)$  satisfies the double limit condition, then every sequence in  $H$  has a subsequence that converges pointwise on  $K$ , to a continuous function.*

*Proof.* By Theorem A.1.3, the set  $H$  is relatively weakly compact. By Fact A.1.4, every sequence in  $H$  has a subsequence converging weakly to some  $f \in C(K)$ . Point evaluations are bounded linear functionals, so the subsequence converges pointwise to  $f$ .  $\odot$

We now connect the criterion with the spaces of types of Section 1.2. For  $\varphi: A \times B \rightarrow [0, 1]$ , recall the evaluation maps  $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$ , given by  $e_b(p) = p(b)$ , and  $e_a: \text{RS}(\varphi) \rightarrow [0, 1]$ , given by  $e_a(q) = q(a)$ , of Subsection 1.2.4.

**Proposition A.1.6.** *Let  $\varphi: A \times B \rightarrow [0, 1]$  be stable. Then  $\{e_b : b \in B\}$  is relatively weakly compact in  $C(\text{LS}(\varphi))$ , and  $\{e_a : a \in A\}$  is relatively weakly compact in  $C(\text{RS}(\varphi))$ . More precisely, the pointwise closure of  $\{e_b : b \in B\}$  consists of the functions  $\hat{\varphi}(\cdot, q)$  with  $q \in \text{RS}(\varphi)$ , where  $\hat{\varphi}$  is the function of Proposition 1.2.8.*

*Proof.* By (1.2) with a constant family, we have  $e_b = \hat{\varphi}(\cdot, \text{rtp}_\varphi(b))$ . If  $q = \lim_{j, \nu} \text{rtp}_\varphi(b_j)$ , then by continuity of  $\hat{\varphi}(p, \cdot)$  we get  $\lim_{j, \nu} e_{b_j}(p) = \hat{\varphi}(p, q)$  for every  $p$ . So every pointwise limit of functions  $e_b$  is of the form  $\hat{\varphi}(\cdot, q)$ , and conversely every such function is a pointwise limit of this kind. These functions are continuous on  $\text{LS}(\varphi)$  by separate continuity. Hence condition (2) of Theorem A.1.3 holds, and  $\{e_b : b \in B\}$  is relatively weakly compact. The statement for  $\{e_a : a \in A\}$  is proved symmetrically. ☺

Proposition A.1.6 places the type spaces of a stable function among the most rigid compact spaces of topology. Recall that a compact space is an *Eberlein compactum* if the space is homeomorphic to a weakly compact subset of a Banach space with the weak topology.

**Corollary A.1.7.** *Let  $\varphi : A \times B \rightarrow [0, 1]$  be stable. Then  $\text{RS}(\varphi)$  and  $\text{LS}(\varphi)$  are Eberlein compacta. In particular, every right  $\varphi$ -type is the limit of a sequence of realized right types, and likewise for left types.*

*Proof.* Let  $H$  be the pointwise closure of  $\{e_b : b \in B\}$  in  $\mathbb{R}^{\text{LS}(\varphi)}$ . By Proposition A.1.6, the weak closure  $W$  of  $\{e_b : b \in B\}$  in  $C(\text{LS}(\varphi))$  is weakly compact. The identity map from  $W$  with the weak topology to  $W$  with the coarser pointwise topology is a continuous bijection from a compact space to a Hausdorff space, so the two topologies agree on  $W$ . In particular, the set  $W$  is pointwise compact, hence pointwise closed, so  $H \subseteq W$ ; and  $W \subseteq H$  because the weak topology is finer than the pointwise topology. Thus  $H = W$ , and on  $H$  the weak and pointwise topologies agree. The map  $q \mapsto \hat{\varphi}(\cdot, q)$  from  $\text{RS}(\varphi)$  onto  $H$  is continuous for the pointwise topology, by the separate continuity of  $\hat{\varphi}$ , and injective, because  $\hat{\varphi}(\text{ltp}_\varphi(a), q) = q(a)$  for every  $a \in A$ . A continuous bijection from a compact space to a Hausdorff space is a homeomorphism, so  $\text{RS}(\varphi)$  is homeomorphic to the weakly compact set  $H$ . The same argument, with the roles of  $A$  and  $B$  exchanged, applies to  $\text{LS}(\varphi)$ . Finally, the set  $H$  is a pointwise compact subset of  $C(\text{LS}(\varphi))$ , hence angelic (see the discussion before Fact A.1.4), so each point of  $H$  is the limit of a sequence from  $\{e_b : b \in B\}$ ; through the homeomorphism, each  $q \in \text{RS}(\varphi)$  is the limit of a sequence of realized types  $\text{rtp}_\varphi(b_n)$ . ☺

Sequences are a genuine restriction. In the unstable structure  $(\omega_1, <)$  with  $\varphi(x, y)$  the formula  $x < y$ , the left type  $p$  given by  $p(\beta) = 0$  for every  $\beta < \omega_1$  (an element above every countable ordinal) is a limit of realized types along an ultrafilter, but not along any sequence: if  $\alpha_n < \omega_1$  for every  $n$ , then  $\alpha = \sup_n \alpha_n < \omega_1$ , and the realized types  $\text{ltp}_\varphi(\alpha_n)$  all take the value 1 at  $\alpha + 1$ , so no subsequence of them converges to  $p$ .

## A.2. Historical remarks

✂ **Section A.1.** Theorem A.1.3 is from Grothendieck's 1952 paper [44], announced in [18], which also proves that pointwise compact subsets of  $C(K)$  are angelic. The Eberlein–Šmulian theorem is due to Šmulian [16] and Eberlein [17]. Later proofs and extensions of the criterion, by Namioka, Pták, Young, Simons, Pryce and König, are listed in the timeline in the historical remarks on Subsection 1.2.1. The reading of the criterion as a statement about stable functions, which is the point of view of Definition A.1.1 and Proposition A.1.6, is due to Ben Yaacov [42] and Pillay [21] for formulas, following Iovino [49] and Krivine–Maurey [51].





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